

14

Boundary-Value Problems in Other Coordinate Systems

EXERCISES 14.1

Problems in Polar Coordinates

1. We have

$$A_0 = \frac{1}{2\pi} \int_0^\pi u_0 d\theta = \frac{u_0}{2}$$

$$A_n = \frac{1}{\pi} \int_0^\pi u_0 \cos n\theta d\theta = 0$$

$$B_n = \frac{1}{\pi} \int_0^\pi u_0 \sin n\theta d\theta = \frac{u_0}{n\pi} [1 - (-1)^n]$$

and so

$$u(r, \theta) = \frac{u_0}{2} + \frac{u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} r^n \sin n\theta.$$

2. We have

$$A_0 = \frac{1}{2\pi} \int_0^\pi \theta d\theta + \frac{1}{2\pi} \int_\pi^{2\pi} (\pi - \theta) d\theta = 0$$

$$A_n = \frac{1}{\pi} \int_0^\pi \theta \cos n\theta d\theta + \frac{1}{\pi} \int_\pi^{2\pi} (\pi - \theta) \cos n\theta d\theta = \frac{2}{n^2\pi} [(-1)^n - 1]$$

$$B_n = \frac{1}{\pi} \int_0^\pi \theta \sin n\theta d\theta + \frac{1}{\pi} \int_\pi^{2\pi} (\pi - \theta) \sin n\theta d\theta = \frac{1}{n} [1 - (-1)^n]$$

and so

$$u(r, \theta) = \sum_{n=1}^{\infty} r^n \left[\frac{(-1)^n - 1}{n^2\pi} \cos n\theta + \frac{1 - (-1)^n}{n} \sin n\theta \right].$$

3. We have

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} (2\pi\theta - \theta^2) d\theta = \frac{2\pi^2}{3}$$

$$A_n = \frac{1}{\pi} \int_0^{2\pi} (2\pi\theta - \theta^2) \cos n\theta d\theta = -\frac{4}{n^2}$$

$$B_n = \frac{1}{\pi} \int_0^{2\pi} (2\pi\theta - \theta^2) \sin n\theta d\theta = 0$$

and so

$$u(r, \theta) = \frac{2\pi^2}{3} - 4 \sum_{n=1}^{\infty} \frac{r^n}{n^2} \cos n\theta.$$

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4. We have

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^{2\pi} \theta \, d\theta = \pi \\ A_n &= \frac{1}{\pi} \int_0^{2\pi} \theta \cos n\theta \, d\theta = 0 \\ B_n &= \frac{1}{\pi} \int_0^{2\pi} \theta \sin n\theta \, d\theta = -\frac{2}{n} \end{aligned}$$

and so

$$u(r, \theta) = \pi - 2 \sum_{n=1}^{\infty} \frac{r^n}{n} \sin n\theta.$$

5. As in Example 1 in the text we have $R(r) = c_3 r^n + c_4 r^{-n}$. In order that the solution be bounded as $r \rightarrow \infty$ we must define $c_3 = 0$. Hence

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^{-n} (A_n \cos n\theta + B_n \sin n\theta)$$

where

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \, d\theta \\ A_n &= \frac{c^n}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta \, d\theta \\ B_n &= \frac{c^n}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta \, d\theta. \end{aligned}$$

6. Using the same reasoning as in Example 1 in the text we obtain

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n (A_n \cos n\theta + B_n \sin n\theta).$$

The boundary condition at $r = c$ implies

$$f(\theta) = \sum_{n=1}^{\infty} nc^{n-1} (A_n \cos n\theta + B_n \sin n\theta).$$

Since this condition does not determine A_0 , it is an arbitrary constant. However, to be a full Fourier series on $[0, 2\pi]$ we must require that $f(\theta)$ satisfy the condition $A_0 = a_0/2 = 0$ or $\int_0^{2\pi} f(\theta) \, d\theta = 0$. If this integral were not 0, then the series for $f(\theta)$ would contain a nonzero constant, which it obviously does not. With this as a necessary compatibility condition we can then make the identifications

$$nc^{n-1}A_n = a_n \quad \text{and} \quad nc^{n-1}B_n = b_n$$

or

$$A_n = \frac{1}{nc^{n-1}\pi} \int_0^{2\pi} f(\theta) \cos n\theta \, d\theta \quad \text{and} \quad B_n = \frac{1}{nc^{n-1}\pi} \int_0^{2\pi} f(\theta) \sin n\theta \, d\theta.$$

7. Proceeding as in Example 1 in the text and again using the periodicity of $u(r, \theta)$, we have

$$\Theta(\theta) = c_1 \cos \alpha \theta + c_2 \sin \alpha \theta$$

where $\alpha = n$ for $n = 0, 1, 2, \dots$. Then

$$R(r) = c_3 r^n + c_4 r^{-n}.$$

[We do not have $c_4 = 0$ in this case since $0 < a \leq r$.] Since $u(b, \theta) = 0$ we have

$$u(r, \theta) = A_0 \ln \frac{r}{b} + \sum_{n=1}^{\infty} \left[\left(\frac{b}{r} \right)^n - \left(\frac{r}{b} \right)^n \right] [A_n \cos n\theta + B_n \sin n\theta].$$

From

$$u(a, \theta) = f(\theta) = A_0 \ln \frac{a}{b} + \sum_{n=1}^{\infty} \left[\left(\frac{b}{a} \right)^n - \left(\frac{a}{b} \right)^n \right] [A_n \cos n\theta + B_n \sin n\theta]$$

we find

$$A_0 \ln \frac{a}{b} = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta,$$

$$\left[\left(\frac{b}{a} \right)^n - \left(\frac{a}{b} \right)^n \right] A_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta d\theta,$$

and

$$\left[\left(\frac{b}{a} \right)^n - \left(\frac{a}{b} \right)^n \right] B_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta d\theta.$$

8. Substituting $u(r, \theta) = v(r, \theta) + \psi(r)$ into the partial differential equation we obtain

$$\frac{\partial^2 v}{\partial r^2} + \psi''(r) + \frac{1}{r} \left[\frac{\partial v}{\partial r} + \psi'(r) \right] + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} = 0.$$

This equation will be homogeneous provided

$$\psi''(r) + \frac{1}{r} \psi'(r) = 0 \quad \text{or} \quad r^2 \psi''(r) + r \psi'(r) = 0.$$

The general solution of this Cauchy-Euler differential equation is

$$\psi(r) = c_1 + c_2 \ln r.$$

From

$$u_0 = u(a, \theta) = v(a, \theta) + \psi(a) \quad \text{and} \quad u_1 = u(b, \theta) = v(b, \theta) + \psi(b)$$

we see that in order for the boundary values $v(a, \theta)$ and $v(b, \theta)$ to be 0 we need $\psi(a) = u_0$ and $\psi(b) = u_1$. From this we have

$$\psi(a) = c_1 + c_2 \ln a = u_0$$

$$\psi(b) = c_1 + c_2 \ln b = u_1.$$

Solving for c_1 and c_2 we obtain

$$c_1 = \frac{u_1 \ln a - u_0 \ln b}{\ln(a/b)} \quad \text{and} \quad c_2 = \frac{u_0 - u_1}{\ln(a/b)}.$$

Then

$$\psi(r) = \frac{u_1 \ln a - u_0 \ln b}{\ln(a/b)} + \frac{u_0 - u_1}{\ln(a/b)} \ln r = \frac{u_0 \ln(r/b) - u_1 \ln(r/a)}{\ln(a/b)}.$$

From Problem 7 with $f(\theta) = 0$ we see that the solution of

$$\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} = 0, \quad 0 < \theta < 2\pi, \quad a < r < b,$$

$$v(a, \theta) = 0, \quad v(b, \theta) = 0, \quad 0 < \theta < 2\pi$$

is $v(r, \theta) = 0$. Thus the steady-state temperature of the ring is

$$u(r, \theta) = v(r, \theta) + \psi(r) = \frac{u_0 \ln(r/b) - u_1 \ln(r/a)}{\ln(a/b)}.$$

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9. This is similar to the solution to Problem 7 above. When $n = 0$, $\Theta(\theta) = c_5\theta + c_6$ and $R(r) = c_7 + c_8 \ln r$. Periodicity implies $c_5 = 0$ and the insulation condition at $r = a$ implies $c_8 = 0$. Thus, we take $u_0 = A_0 = c_6c_7$. Then, for $n = 1, 2, 3, \dots$, $\Theta(\theta) = c_1 \cos n\theta + c_2 \sin n\theta$ and $R(r) = c_3r^n + c_4r^{-n}$. From $R'(a) = 0$ we get $c_3na^{n-1} - c_4na^{-n-1} = 0$, which implies $c_4 = c_3a^{2n}$. Then

$$R(r) = c_3(r^n + a^{2n}r^{-n}) = c_3 \frac{r^{2n} + a^{2n}}{r^n}$$

and

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} \frac{r^{2n} + a^{2n}}{r^n} (A_n \cos n\theta + B_n \sin n\theta).$$

Taking $r = b$ we have

$$f(\theta) = A_0 + \sum_{n=1}^{\infty} \frac{b^{2n} + a^{2n}}{b^n} (A_n \cos n\theta + B_n \sin n\theta),$$

which implies

$$A_0 = \frac{a_0}{2} = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta$$

and

$$\frac{b^{2n} + a^{2n}}{b^n} A_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta d\theta \quad \text{and} \quad \frac{b^{2n} + a^{2n}}{b^n} B_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin 2n\theta d\theta.$$

Hence

$$A_n = \frac{b^n}{\pi(a^{2n} + b^{2n})} \int_0^{2\pi} f(\theta) \cos n\theta d\theta \quad \text{and} \quad B_n = \frac{b^n}{\pi(a^{2n} + b^{2n})} \int_0^{2\pi} f(\theta) \sin n\theta d\theta.$$

10. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < \theta < \frac{\pi}{2}, \quad 0 < r < c,$$

$$u(c, \theta) = f(\theta), \quad 0 < \theta < \frac{\pi}{2},$$

$$u(r, 0) = 0, \quad u(r, \pi/2) = 0, \quad 0 < r < c.$$

Proceeding as in Example 1 in the text we obtain the separated differential equations

$$r^2 R'' + rR' - \lambda R = 0$$

$$\Theta'' + \lambda \Theta = 0.$$

Taking $\lambda = \alpha^2$ the solutions are

$$\Theta(\theta) = c_1 \cos \alpha\theta + c_2 \sin \alpha\theta$$

$$R(r) = c_3 r^\alpha + c_4 r^{-\alpha}.$$

Since we want $R(r)$ to be bounded as $r \rightarrow 0$ we require $c_4 = 0$. Applying the boundary conditions $\Theta(0) = 0$ and $\Theta(\pi/2) = 0$ we find that $c_1 = 0$ and $\alpha = 2n$ for $n = 1, 2, 3, \dots$. Therefore

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{2n} \sin 2n\theta.$$

From

$$u(c, \theta) = f(\theta) = \sum_{n=1}^{\infty} A_n c^n \sin 2n\theta$$

we find

$$A_n = \frac{4}{\pi c^{2n}} \int_0^{\pi/2} f(\theta) \sin 2n\theta d\theta.$$

11. Referring to the solution of Problem 10 above we have

$$\Theta(\theta) = c_1 \cos \alpha \theta + c_2 \sin \alpha \theta$$

$$R(r) = c_3 r^\alpha.$$

Applying the boundary conditions $\Theta'(0) = 0$ and $\Theta'(\pi/2) = 0$ we find that $c_2 = 0$ and $\alpha = 2n$ for $n = 0, 1, 2, \dots$. Therefore

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} A_n r^{2n} \cos 2n\theta.$$

From

$$u(c, \theta) = \begin{cases} 1, & 0 < \theta < \pi/4 \\ 0, & \pi/4 < \theta < \pi/2 \end{cases} = A_0 + \sum_{n=1}^{\infty} A_n c^{2n} \cos 2n\theta$$

we find

$$A_0 = \frac{1}{\pi/2} \int_0^{\pi/4} d\theta = \frac{1}{2}$$

and

$$c^{2n} A_n = \frac{2}{\pi/2} \int_0^{\pi/4} \cos 2n\theta d\theta = \frac{2}{n\pi} \sin \frac{n\pi}{2}.$$

Thus

$$u(r, \theta) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{2} \left(\frac{r}{c}\right)^{2n} \cos 2n\theta.$$

12. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < \theta < \pi/4, \quad r > 0$$

$$u(r, 0) = 0, \quad r > 0$$

$$u(r, \pi/4) = 30, \quad r > 0.$$

Proceeding as in Example 1 in the text we find the separated ordinary differential equations to be

$$r^2 R'' + r R' - \lambda R = 0$$

$$\Theta'' + \lambda \Theta = 0.$$

With $\lambda = \alpha^2 > 0$ the corresponding general solutions are

$$R(r) = c_1 r^\alpha + c_2 r^{-\alpha}$$

$$\Theta(\theta) = c_3 \cos \alpha \theta + c_4 \sin \alpha \theta.$$

The condition $\Theta(0) = 0$ implies $c_3 = 0$ so that $\Theta = c_4 \sin \alpha \theta$. Now, in order that the temperature be bounded as $r \rightarrow \infty$ we define $c_1 = 0$. Similarly, in order that the temperature be bounded as $r \rightarrow 0$ we are forced to define $c_2 = 0$. Thus $R(r) = 0$ and so no nontrivial solution exists for $\lambda > 0$. For $\lambda = 0$ the separated differential equations are

$$r^2 R'' + r R' = 0 \quad \text{and} \quad \Theta'' = 0.$$

Solutions of these latter equations are

$$R(r) = c_1 + c_2 \ln r \quad \text{and} \quad \Theta(\theta) = c_3 \theta + c_4.$$

$\Theta(0) = 0$ still implies $c_4 = 0$, whereas boundedness as $r \rightarrow 0$ demands $c_2 = 0$. Thus, a product solution is

$$u = c_1 c_3 \theta = A \theta.$$

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From $u(r, \pi/4) = 0$ we obtain $A = 120/\pi$. Thus, a solution to the problem is

$$u(r, \theta) = \frac{120}{\pi} \theta.$$

13. We solve

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad 0 < \theta < \pi, \quad a < r < b, \\ u(a, \theta) &= \theta(\pi - \theta), \quad u(b, \theta) = 0, \quad 0 < \theta < \pi, \\ u(r, 0) &= 0, \quad u(r, \pi) = 0, \quad a < r < b. \end{aligned}$$

Proceeding as in Example 1 in the text we obtain the separated differential equations

$$\begin{aligned} r^2 R'' + rR' - \lambda R &= 0 \\ \Theta'' + \lambda \Theta &= 0. \end{aligned}$$

Taking $\lambda = \alpha^2$ the solutions are

$$\begin{aligned} \Theta(\theta) &= c_1 \cos \alpha \theta + c_2 \sin \alpha \theta \\ R(r) &= c_3 r^\alpha + c_4 r^{-\alpha}. \end{aligned}$$

Applying the boundary conditions $\Theta(0) = 0$ and $\Theta(\pi) = 0$ we find that $c_1 = 0$ and $\alpha = n$ for $n = 1, 2, 3, \dots$. The boundary condition $R(b) = 0$ gives

$$c_3 b^n + c_4 b^{-n} = 0 \quad \text{and} \quad c_4 = -c_3 b^{2n}.$$

Then

$$R(r) = c_3 \left(r^n - \frac{b^{2n}}{r^n} \right) = c_3 \left(\frac{r^{2n} - b^{2n}}{r^n} \right)$$

and

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n \left(\frac{r^{2n} - b^{2n}}{r^n} \right) \sin n\theta.$$

From

$$u(a, \theta) = \theta(\pi - \theta) = \sum_{n=1}^{\infty} A_n \left(\frac{a^{2n} - b^{2n}}{a^n} \right) \sin n\theta$$

we find

$$A_n \left(\frac{a^{2n} - b^{2n}}{a^n} \right) = \frac{2}{\pi} \int_0^\pi (\theta\pi - \theta^2) \sin n\theta \, d\theta = \frac{4}{n^3 \pi} [1 - (-1)^n].$$

Thus

$$u(r, \theta) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \frac{r^{2n} - b^{2n}}{a^{2n} - b^{2n}} \left(\frac{a}{r} \right)^n \sin n\theta.$$

14. Letting $u(r, \theta) = v(r, \theta) + \psi(\theta)$ we obtain $\psi''(\theta) = 0$ and so $\psi(\theta) = c_1 \theta + c_2$. From $\psi(0) = 0$ and $\psi(\pi) = u_0$ we find, in turn, $c_2 = 0$ and $c_1 = u_0/\pi$. Therefore $\psi(\theta) = \frac{u_0}{\pi} \theta$. Now $u(1, \theta) = v(1, \theta) + \psi(\theta)$ so that $v(1, \theta) = u_0 - \frac{u_0}{\pi} \theta$. From

$$v(r, \theta) = \sum_{n=1}^{\infty} A_n r^n \sin n\theta \quad \text{and} \quad v(1, \theta) = \sum_{n=1}^{\infty} A_n \sin n\theta$$

we obtain

$$A_n = \frac{2}{\pi} \int_0^\pi \left(u_0 - \frac{u_0}{\pi} \theta \right) \sin n\theta \, d\theta = \frac{2u_0}{\pi n}.$$

Thus

$$u(r, \theta) = \frac{u_0}{\pi} \theta + \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{r^n}{n} \sin n\theta.$$

15. We solve

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad 0 < \theta < \pi, \quad 0 < r < 2, \\ u(2, \theta) &= \begin{cases} u_0, & 0 < \theta < \pi/2 \\ 0, & \pi/2 < \theta < \pi \end{cases} \\ \frac{\partial u}{\partial \theta} \Big|_{\theta=0} &= 0, \quad \frac{\partial u}{\partial \theta} \Big|_{\theta=\pi} = 0, \quad 0 < r < 2. \end{aligned}$$

Proceeding as in Example 1 in the text we obtain the separated differential equations

$$\begin{aligned} r^2 R'' + rR' - \lambda R &= 0 \\ \Theta'' + \lambda \Theta &= 0. \end{aligned}$$

Taking $\lambda = \alpha^2$ the solutions are

$$\begin{aligned} \Theta(\theta) &= c_1 \cos \alpha \theta + c_2 \sin \alpha \theta \\ R(r) &= c_3 r^\alpha + c_4 r^{-\alpha}. \end{aligned}$$

Applying the boundary conditions $\Theta'(0) = 0$ and $\Theta'(\pi) = 0$ we find that $c_2 = 0$ and $\alpha = n$ for $n = 0, 1, 2, \dots$. Since we want $R(r)$ to be bounded as $r \rightarrow 0$ we require $c_4 = 0$. Thus

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} A_n r^n \cos n\theta.$$

From

$$u(2, \theta) = \begin{cases} u_0, & 0 < \theta < \pi/2 \\ 0, & \pi/2 < \theta < \pi \end{cases} = A_0 + \sum_{n=1}^{\infty} A_n 2^n \cos n\theta$$

we find

$$A_0 = \frac{1}{2} \frac{2}{\pi} \int_0^{\pi/2} u_0 d\theta = \frac{u_0}{2}$$

and

$$2^n A_n = \frac{2u_0}{\pi} \int_0^{\pi/2} \cos n\theta d\theta = \frac{2u_0}{\pi} \frac{\sin n\pi/2}{n}.$$

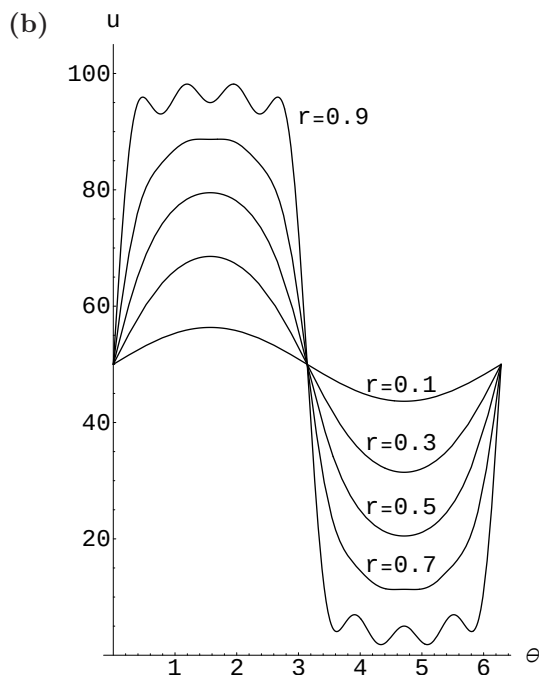
Therefore

$$u(r, \theta) = \frac{u_0}{2} + \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \left(\sin \frac{n\pi}{2} \right) \left(\frac{r}{2} \right)^n \cos n\theta.$$

16. (a) From Problem 1 in this section, with $u_0 = 100$,

$$u(r, \theta) = 50 + \frac{100}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} r^n \sin n\theta.$$

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- (c) We could use S_5 from part (b) of this problem to compute the approximations, but in a CAS it is just as easy to compute the sum with a much larger number of terms, thereby getting greater accuracy. In this case we use partial sums including the term with r^{99} to find

$u(0.9, 1.3) \approx 96.5268$	$u(0.9, 2\pi - 1.3) \approx 3.4731$
$u(0.7, 2) \approx 87.871$	$u(0.7, 2\pi - 2) \approx 12.129$
$u(0.5, 3.5) \approx 36.0744$	$u(0.5, 2\pi - 3.5) \approx 63.9256$
$u(0.3, 4) \approx 35.2674$	$u(0.3, 2\pi - 4) \approx 64.7326$
$u(0.1, 5.5) \approx 45.4934$	$u(0.1, 2\pi - 5.5) \approx 54.5066$

- (d) At the center of the plate $u(0, 0) = 50$. From the graphs in part (b) we observe that the solution curves are symmetric about the point $(\pi, 50)$. In part (c) we observe that the horizontal pairs add up to 100, and hence average 50. This is consistent with the observation about part (b), so it is appropriate to say the average temperature in the plate is 50° .

17. Let u_1 be the solution of the boundary-value problem

$$\frac{\partial^2 u_1}{\partial r^2} + \frac{1}{r} \frac{\partial u_1}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u_1}{\partial \theta^2} = 0, \quad 0 < \theta < 2\pi, \quad a < r < b$$

$$u_1(a, \theta) = f(\theta), \quad 0 < \theta < 2\pi$$

$$u_1(b, \theta) = 0, \quad 0 < \theta < 2\pi,$$

and let u_2 be the solution of the boundary-value problem

$$\frac{\partial^2 u_2}{\partial r^2} + \frac{1}{r} \frac{\partial u_2}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u_2}{\partial \theta^2} = 0, \quad 0 < \theta < 2\pi, \quad a < r < b$$

$$u_2(a, \theta) = 0, \quad 0 < \theta < 2\pi$$

$$u_2(b, \theta) = g(\theta), \quad 0 < \theta < 2\pi.$$

Each of these problems can be solved using the methods shown in Problem 7 of this section. Now if $u(r, \theta) = u_1(r, \theta) + u_2(r, \theta)$, then

$$u(a, \theta) = u_1(a, \theta) + u_2(a, \theta) = f(\theta)$$

$$u(b, \theta) = u_1(b, \theta) + u_2(b, \theta) = g(\theta)$$

and $u(r, \theta)$ will be the steady-state temperature of the circular ring with boundary conditions $u(a, \theta) = f(\theta)$ and $u(b, \theta) = g(\theta)$.

EXERCISES 14.2

Problems in Cylindrical Coordinates

- Referring to the solution of Example 1 in the text we have

$$R(r) = c_1 J_0(\alpha_n r) \quad \text{and} \quad T(t) = c_3 \cos a \alpha_n t + c_4 \sin a \alpha_n t$$

where the α_n are the positive roots of $J_0(\alpha c) = 0$. Now, the initial condition $u(r, 0) = R(r)T(0) = 0$ implies $T(0) = 0$ and so $c_3 = 0$. Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n \sin a \alpha_n t J_0(\alpha_n r) \quad \text{and} \quad \frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} a \alpha_n A_n \cos a \alpha_n t J_0(\alpha_n r).$$

From

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = 1 = \sum_{n=1}^{\infty} a \alpha_n A_n J_0(\alpha_n r)$$

we find

$$\begin{aligned} a \alpha_n A_n &= \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^c r J_0(\alpha_n r) dr && \boxed{x = \alpha_n r, \quad dx = \alpha_n dr} \\ &= \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^{\alpha_n c} \frac{1}{\alpha_n^2} x J_0(x) dx \\ &= \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^{\alpha_n c} \frac{1}{\alpha_n^2} \frac{d}{dx} [x J_1(x)] dx && \boxed{\text{see (4) of Section 12.6 in text}} \\ &= \frac{2}{c^2 \alpha_n^2 J_1^2(\alpha_n c)} x J_1(x) \Big|_0^{\alpha_n c} = \frac{2}{c \alpha_n J_1(\alpha_n c)}. \end{aligned}$$

Then

$$A_n = \frac{2}{a c \alpha_n^2 J_1(\alpha_n c)}$$

and

$$u(r, t) = \frac{2}{a c} \sum_{n=1}^{\infty} \frac{J_0(\alpha_n r)}{\alpha_n^2 J_1(\alpha_n c)} \sin a \alpha_n t.$$

- From Example 1 in the text we have $B_n = 0$ and

$$A_n = \frac{2}{J_1^2(\alpha_n)} \int_0^1 r(1 - r^2) J_0(\alpha_n r) dr.$$

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From Problem 10, Exercises 12.6 we obtained $A_n = \frac{4J_2(\alpha_n)}{\alpha_n^2 J_1^2(\alpha_n)}$. Thus

$$u(r, t) = 4 \sum_{n=1}^{\infty} \frac{J_2(\alpha_n)}{J_1^2(\alpha_n)} \cos \alpha_n t J_0(\alpha_n r).$$

3. Referring to Example 2 in the text we have

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)$$

$$Z(z) = c_3 \cosh \alpha z + c_4 \sinh \alpha z$$

where $c_2 = 0$ and $J_0(2\alpha) = 0$ defines the positive eigenvalues $\lambda_n = \alpha_n^2$. From $Z(4) = 0$ we obtain

$$c_3 \cosh 4\alpha_n + c_4 \sinh 4\alpha_n = 0 \quad \text{or} \quad c_4 = -c_3 \frac{\cosh 4\alpha_n}{\sinh 4\alpha_n}.$$

Then

$$\begin{aligned} Z(z) &= c_3 \left[\cosh \alpha_n z - \frac{\cosh 4\alpha_n}{\sinh 4\alpha_n} \sinh \alpha_n z \right] = c_3 \frac{\sinh 4\alpha_n \cosh \alpha_n z - \cosh 4\alpha_n \sinh \alpha_n z}{\sinh 4\alpha_n} \\ &= c_3 \frac{\sinh \alpha_n (4 - z)}{\sinh 4\alpha_n} \end{aligned}$$

and

$$u(r, z) = \sum_{n=1}^{\infty} A_n \frac{\sinh \alpha_n (4 - z)}{\sinh 4\alpha_n} J_0(\alpha_n r).$$

From

$$u(r, 0) = u_0 = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)$$

we obtain

$$A_n = \frac{2u_0}{4J_1^2(2\alpha_n)} \int_0^2 r J_0(\alpha_n r) dr = \frac{u_0}{\alpha_n J_1(2\alpha_n)}.$$

Thus the temperature in the cylinder is

$$u(r, z) = u_0 \sum_{n=1}^{\infty} \frac{\sinh \alpha_n (4 - z) J_0(\alpha_n r)}{\alpha_n \sinh 4\alpha_n J_1(2\alpha_n)}.$$

4. (a) The boundary condition $u_r(2, z) = 0$ implies $R'(2) = 0$ or $J'_0(2\alpha) = 0$. Thus $\alpha = 0$ is also an eigenvalue and the separated equations are in this case $rR'' + R' = 0$ and $z'' = 0$. The solutions of these equations are then

$$R(r) = c_1 + c_2 \ln r, \quad Z(z) = c_3 z + c_4.$$

Now $Z(0) = 0$ yields $c_4 = 0$ and the implicit condition that the temperature is bounded as $r \rightarrow 0$ demands that we define $c_2 = 0$. Thus we have

$$u(r, z) = A_1 z + \sum_{n=2}^{\infty} A_n \sinh \alpha_n z J_0(\alpha_n r). \quad (1)$$

At $z = 4$ we obtain

$$f(r) = 4A_1 + \sum_{n=2}^{\infty} A_n \sinh 4\alpha_n J_0(\alpha_n r).$$

Thus from (17) and (18) of Section 12.6 in the text we can write with $b = 2$,

$$A_1 = \frac{1}{8} \int_0^2 r f(r) dr \quad (2)$$

$$A_n = \frac{1}{2 \sinh 4\alpha_n J_0^2(2\alpha_n)} \int_0^2 r f(r) J_0(\alpha_n r) dr. \quad (3)$$

A solution of the problem consists of the series (1) with coefficients A_1 and A_n defined in (2) and (3), respectively.

(b) When $f(r) = u_0$ we get $A_1 = u_0/4$ and

$$A_n = \frac{u_0 J_1(2\alpha_n)}{\alpha_n \sinh 4\alpha_n J_0^2(2\alpha_n)} = 0$$

since $J'_0(2\alpha) = 0$ is equivalent to $J_1(2\alpha) = 0$. A solution of the problem is then $u(r, z) = \frac{u_0}{4} z$.

5. Letting $u(r, t) = R(r)T(t)$ and separating variables we obtain

$$\frac{R'' + \frac{1}{r}R'}{R} = \frac{T'}{kT} = -\lambda \quad \text{and} \quad R'' + \frac{1}{r}R' + \lambda R = 0, \quad T' + \lambda kT = 0.$$

From the last equation we find $T(t) = e^{-\lambda kt}$. If $\lambda < 0$, $T(t)$ increases without bound as $t \rightarrow \infty$. Thus we assume $\lambda = \alpha^2 > 0$. Now

$$R'' + \frac{1}{r}R' + \alpha^2 R = 0$$

is a parametric Bessel equation with solution

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r).$$

Since Y_0 is unbounded as $r \rightarrow 0$ we take $c_2 = 0$. Then $R(r) = c_1 J_0(\alpha r)$ and the boundary condition $u(c, t) = R(c)T(t) = 0$ implies $J_0(\alpha c) = 0$. This latter equation defines the positive eigenvalues $\lambda_n = \alpha_n^2$. Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) e^{-\alpha_n^2 kt}.$$

From

$$u(r, 0) = f(r) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)$$

we find

$$A_n = \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^c r J_0(\alpha_n r) f(r) dr, \quad n = 1, 2, 3, \dots$$

6. If the edge $r = c$ is insulated we have the boundary condition $u_r(c, t) = 0$. Referring to the solution of Problem 5 above we have

$$R'(c) = \alpha c_1 J'_0(\alpha c) = 0$$

which defines an eigenvalue $\lambda = \alpha^2 = 0$ and positive eigenvalues $\lambda_n = \alpha_n^2$. Thus

$$u(r, t) = A_0 + \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) e^{-\alpha_n^2 kt}.$$

From

$$u(r, 0) = f(r) = A_0 + \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)$$

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we find

$$A_0 = \frac{2}{c^2} \int_0^c r f(r) dr$$

$$A_n = \frac{2}{c^2 J_0^2(\alpha_n c)} \int_0^c r J_0(\alpha_n r) f(r) dr.$$

7. Referring to Problem 5 above we have $T(t) = e^{-\lambda k t}$ and $R(r) = c_1 J_0(\alpha r)$. The boundary condition $hu(1, t) + u_r(1, t) = 0$ implies $hJ_0(\alpha) + \alpha J_0'(\alpha) = 0$ which defines positive eigenvalues $\lambda_n = \alpha_n^2$. Now

$$u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) e^{-\alpha_n^2 k t}$$

where

$$A_n = \frac{2\alpha_n^2}{(\alpha_n^2 + h^2) J_0^2(\alpha_n)} \int_0^1 r J_0(\alpha_n r) f(r) dr.$$

8. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial z^2} = 0, \quad 0 < r < 1, \quad z > 0$$

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} = -hu(1, z), \quad z > 0$$

$$u(r, 0) = u_0, \quad 0 < r < 1.$$

assuming $u = RZ$ we get

$$\frac{R'' + \frac{1}{r}R'}{R} = -\frac{Z''}{Z} = -\lambda$$

and so

$$rR'' + R' + \lambda^2 rR = 0 \quad \text{and} \quad Z'' - \lambda Z = 0.$$

Letting $\lambda = \alpha^2$ we then have

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r) \quad \text{and} \quad Z(z) = c_3 e^{-\alpha z} + c_4 e^{\alpha z}.$$

We use the exponential form of the solution of $Z'' - \alpha^2 Z = 0$ since the domain of the variable z is a semi-infinite interval. As usual we define $c_2 = 0$ since the temperature is surely bounded as $r \rightarrow 0$. Hence $R(r) = c_1 J_0(\alpha r)$. Now the boundary-condition $u_r(1, z) + hu(1, z) = 0$ is equivalent to

$$\alpha J_0'(\alpha) + hJ_0(\alpha) = 0. \quad (4)$$

The eigenvalues α_n are the positive roots of (4) above. Finally, we must now define $c_4 = 0$ since the temperature is also expected to be bounded as $z \rightarrow \infty$. A product solution of the partial differential equation that satisfies the first boundary condition is given by

$$u_n(r, z) = A_n e^{-\alpha_n z} J_0(\alpha_n r).$$

Therefore

$$u(r, z) = \sum_{n=1}^{\infty} A_n e^{-\alpha_n z} J_0(\alpha_n r)$$

is another formal solution. At $z = 0$ we have $u_0 = A_n J_0(\alpha_n r)$. In view of (4) above we use equations (17) and (18) of Section 12.6 in the text with the identification $b = 1$:

$$A_n = \frac{2\alpha_n^2}{(\alpha_n^2 + h^2) J_0^2(\alpha_n)} \int_0^1 r J_0(\alpha_n r) u_0 dr$$

$$= \frac{2\alpha_n^2 u_0}{(\alpha_n^2 + h^2) J_0^2(\alpha_n) \alpha_n^2} t J_1(t) \Big|_0^{\alpha_n} = \frac{2\alpha_n u_0 J_1(\alpha_n)}{(\alpha_n^2 + h^2) J_0^2(\alpha_n)}. \quad (5)$$

Since $J'_0 = -J_1$ [see equation (6) of Section 11.5 in the text] it follows from (4) above that $\alpha_n J_1(\alpha_n) = h J_0(\alpha_n)$. Thus (5) above simplifies to

$$A_n = \frac{2u_0 h}{(\alpha_n^2 + h^2) J_0(\alpha_n)}.$$

A solution to the boundary-value problem is then

$$u(r, z) = 2u_0 h \sum_{n=1}^{\infty} \frac{e^{-\alpha_n z}}{(\alpha_n^2 + h^2) J_0(\alpha_n)} J_0(\alpha_n r).$$

9. Substituting $u(r, t) = v(r, t) + \psi(r)$ into the partial differential equation gives

$$\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \psi'' + \frac{1}{r} \psi' = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided $\psi'' + \frac{1}{r} \psi' = 0$ or $\psi(r) = c_1 \ln r + c_2$. Since $\ln r$ is unbounded as $r \rightarrow 0$ we take $c_1 = 0$. Then $\psi(r) = c_2$ and using $u(2, t) = v(2, t) + \psi(2) = 100$ we set $c_2 = \psi(2) = 100$. Therefore $\psi(r) = 100$. Referring to Problem 5 above, the solution of the boundary-value problem

$$\begin{aligned} \frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} &= \frac{\partial v}{\partial t}, \quad 0 < r < 2, \quad t > 0, \\ v(2, t) &= 0, \quad t > 0, \\ v(r, 0) &= u(r, 0) - \psi(r) \end{aligned}$$

is

$$v(r, t) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) e^{-\alpha_n^2 t}$$

where

$$\begin{aligned} A_n &= \frac{2}{2^2 J_1^2(2\alpha_n)} \int_0^2 r J_0(\alpha_n r) [u(r, 0) - \psi(r)] dr \\ &= \frac{1}{2 J_1^2(2\alpha_n)} \left[\int_0^1 r J_0(\alpha_n r) [200 - 100] dr + \int_1^2 r J_0(\alpha_n r) [100 - 100] dr \right] \\ &= \frac{50}{J_1^2(2\alpha_n)} \int_0^1 r J_0(\alpha_n r) dr \quad \boxed{x = \alpha_n r, \quad dx = \alpha_n dr} \\ &= \frac{50}{J_1^2(2\alpha_n)} \int_0^{\alpha_n} \frac{1}{\alpha_n^2} x J_0(x) dx \\ &= \frac{50}{\alpha_n^2 J_1^2(2\alpha_n)} \int_0^{\alpha_n} \frac{d}{dx} [x J_1(x)] dx \quad \boxed{\text{see (5) of Section 12.6 in text}} \\ &= \frac{50}{\alpha_n^2 J_1^2(2\alpha_n)} (x J_1(x)) \Big|_0^{\alpha_n} = \frac{50 J_1(\alpha_n)}{\alpha_n J_1^2(2\alpha_n)}. \end{aligned}$$

Thus

$$u(r, t) = v(r, t) + \psi(r) = 100 + 50 \sum_{n=1}^{\infty} \frac{J_1(\alpha_n) J_0(\alpha_n r)}{\alpha_n J_1^2(2\alpha_n)} e^{-\alpha_n^2 t}.$$

10. Letting $u(r, t) = u(r, t) + \psi(r)$ we obtain $r\psi'' + \psi' = -\beta r$. The general solution of this nonhomogeneous Cauchy-Euler equation is found with the aid of variation of parameters: $\psi = c_1 + c_2 \ln r - \beta r^2/4$. In order that this

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solution be bounded as $r \rightarrow 0$ we define $c_2 = 0$. Using $\psi(1) = 0$ then gives $c_1 = \beta/4$ and so $\psi(r) = \beta(1 - r^2)/4$. Using $v = RT$ we find that a solution of

$$\begin{aligned}\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} &= \frac{\partial v}{\partial t}, \quad 0 < r < 1, \quad t > 0 \\ v(1, t) &= 0, \quad t > 0 \\ v(r, 0) &= -\psi(r), \quad 0 < r < 1\end{aligned}$$

is

$$v(r, t) = \sum_{n=1}^{\infty} A_n e^{-\alpha_n^2 t} J_0(\alpha_n r)$$

where

$$A_n = -\frac{\beta}{4} \frac{2}{J_1^2(\alpha_n)} \int_0^1 r(1 - r^2) J_0(\alpha_n r) dr$$

and the α_n are defined by $J_0(\alpha) = 0$. From the result of Problem 10, Exercises 12.6 (see also Problem 2 of this exercise set) we get

$$A_n = -\frac{\beta J_2(\alpha_n)}{\alpha_n^2 J_1^2(\alpha_n)}.$$

Thus from $u = v + \psi(r)$ it follows that

$$u(r, t) = \frac{\beta}{4}(1 - r^2) - \beta \sum_{n=1}^{\infty} \frac{J_2(\alpha_n)}{\alpha_n^2 J_1^2(\alpha_n)} e^{-\alpha_n^2 t} J_0(\alpha_n r).$$

11. (a) Writing the partial differential equation in the form

$$g \left(x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \right) = \frac{\partial^2 u}{\partial t^2}$$

and separating variables we obtain

$$\frac{xX'' + X'}{X} = \frac{T''}{gT} = -\lambda.$$

Letting $\lambda = \alpha^2$ we obtain

$$xX'' + X' + \alpha^2 X = 0 \quad \text{and} \quad T'' + g\alpha^2 T = 0.$$

Letting $x = \tau^2/4$ in the first equation we obtain $dx/d\tau = \tau/2$ or $d\tau/dx = 2\tau$. Then

$$\frac{dX}{dx} = \frac{dX}{d\tau} \frac{d\tau}{dx} = \frac{2}{\tau} \frac{dX}{d\tau}$$

and

$$\begin{aligned}\frac{d^2 X}{dx^2} &= \frac{d}{dx} \left(\frac{2}{\tau} \frac{dX}{d\tau} \right) = \frac{2}{\tau} \frac{d}{dx} \left(\frac{dX}{d\tau} \right) + \frac{dX}{d\tau} \frac{d}{dx} \left(\frac{2}{\tau} \right) \\ &= \frac{2}{\tau} \frac{d}{d\tau} \left(\frac{dX}{d\tau} \right) \frac{d\tau}{dx} + \frac{dX}{d\tau} \frac{d}{d\tau} \left(\frac{2}{\tau} \right) \frac{d\tau}{dx} = \frac{4}{\tau^2} \frac{d^2 X}{d\tau^2} - \frac{4}{\tau^3} \frac{dX}{d\tau}.\end{aligned}$$

Thus

$$xX'' + X' + \alpha^2 X = \frac{\tau^2}{4} \left(\frac{4}{\tau^2} \frac{d^2 X}{d\tau^2} - \frac{4}{\tau^3} \frac{dX}{d\tau} \right) + \frac{2}{\tau} \frac{dX}{d\tau} + \alpha^2 X = \frac{d^2 X}{d\tau^2} + \frac{1}{\tau} \frac{dX}{d\tau} + \alpha^2 X = 0.$$

This is a parametric Bessel equation with solution

$$X(\tau) = c_1 J_0(\alpha\tau) + c_2 Y_0(\alpha\tau).$$

- (b) To insure a finite solution at $x = 0$ (and thus $\tau = 0$) we set $c_2 = 0$. The condition $u(L, t) = X(L)T(t) = 0$ implies $X|_{x=L} = X|_{\tau=2\sqrt{L}} = c_1 J_0(2\alpha\sqrt{L}) = 0$, which defines positive eigenvalues $\lambda_n = \alpha_n^2$. The solution

of $T'' + g\alpha^2 T = 0$ is

$$T(t) = c_3 \cos(\alpha_n \sqrt{g} t) + c_4 \sin(\alpha_n \sqrt{g} t).$$

The boundary condition $u_t(x, 0) = X(x)T'(0) = 0$ implies $c_4 = 0$. Thus

$$u(\tau, t) = \sum_{n=1}^{\infty} A_n \cos(\alpha_n \sqrt{g} t) J_0(\alpha_n \tau).$$

From

$$u(\tau, 0) = f(\tau^2/4) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n \tau)$$

we find

$$\begin{aligned} A_n &= \frac{2}{(2\sqrt{L})^2 J_1^2(2\alpha_n \sqrt{L})} \int_0^{2\sqrt{L}} \tau J_0(\alpha_n \tau) f(\tau^2/4) d\tau \quad \boxed{v = \tau/2, \quad dv = d\tau/2} \\ &= \frac{1}{2L J_1^2(2\alpha_n \sqrt{L})} \int_0^{\sqrt{L}} 2v J_0(2\alpha_n v) f(v^2) 2dv \\ &= \frac{2}{L J_1^2(2\alpha_n \sqrt{L})} \int_0^{\sqrt{L}} v J_0(2\alpha_n v) f(v^2) dv. \end{aligned}$$

The solution of the boundary-value problem is

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos(\alpha_n \sqrt{g} t) J_0(2\alpha_n \sqrt{x}).$$

12. (a) First we see that

$$\frac{R''\Theta + \frac{1}{r}R'\Theta + \frac{1}{r^2}R\Theta''}{R\Theta} = \frac{T''}{a^2T} = -\lambda.$$

This gives $T'' + a^2\lambda T = 0$ and from

$$\frac{R'' + \frac{1}{r}R' + \lambda R}{-R/r^2} = \frac{\Theta''}{\Theta} = -\nu$$

we get $\Theta'' + \nu\Theta = 0$ and $r^2R'' + rR' + (\lambda r^2 - \nu)R = 0$.

- (b) With $\lambda = \alpha^2$ and $\nu = \beta^2$ the general solutions of the differential equations in part (a) are

$$T = c_1 \cos \alpha \alpha t + c_2 \sin \alpha \alpha t$$

$$\Theta = c_3 \cos \beta \theta + c_4 \sin \beta \theta$$

$$R = c_5 J_\beta(\alpha r) + c_6 Y_\beta(\alpha r).$$

- (c) Implicitly we expect $u(r, \theta, t) = u(r, \theta + 2\pi, t)$ and so Θ must be 2π -periodic. Therefore $\beta = n$, $n = 0, 1, 2, \dots$. The corresponding eigenfunctions are $1, \cos \theta, \cos 2\theta, \dots, \sin \theta, \sin 2\theta, \dots$. Arguing that $u(r, \theta, t)$ is bounded as $r \rightarrow 0$ we then define $c_6 = 0$ and so $R = c_3 J_n(\alpha r)$. But $R(c) = 0$ gives $J_n(\alpha c) = 0$; this equation defines the eigenvalues $\lambda_n = \alpha_n^2$. For each n , $\alpha_{ni} = x_{ni}/c$, $i = 1, 2, 3, \dots$. The corresponding eigenfunctions are $J_n(\lambda_{ni}r) = 0$.

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$$\begin{aligned}
 \text{(d)} \quad u(r, \theta, t) = & \sum_{i=1}^n (A_{0i} \cos a\alpha_{0i}t + B_{0i} \sin a\alpha_{0i}t) J_0(\alpha_{0i}r) \\
 & + \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \left[(A_{ni} \cos a\alpha_{ni}t + B_{ni} \sin a\alpha_{ni}t) \cos n\theta \right. \\
 & \left. + (C_{ni} \cos a\alpha_{ni}t + D_{ni} \sin a\alpha_{ni}t) \sin n\theta \right] J_n(\alpha_{ni}r)
 \end{aligned}$$

13. (a) With $c = 10$ in Example 1 in the text the eigenvalues are $\lambda_n = \alpha_n^2 = x_n^2/100$ where x_n is a positive root of $J_0(x) = 0$. From a CAS we find that $x_1 = 2.4048$, $x_2 = 5.5201$, and $x_3 = 8.6537$, so that the first three eigenvalues are $\lambda_1 = 0.0578$, $\lambda_2 = 0.3047$, and $\lambda_3 = 0.7489$. The corresponding coefficients are

$$A_1 = \frac{2}{100J_1^2(x_1)} \int_0^{10} r J_0(x_1 r/10)(1 - r/10) dr = 0.7845,$$

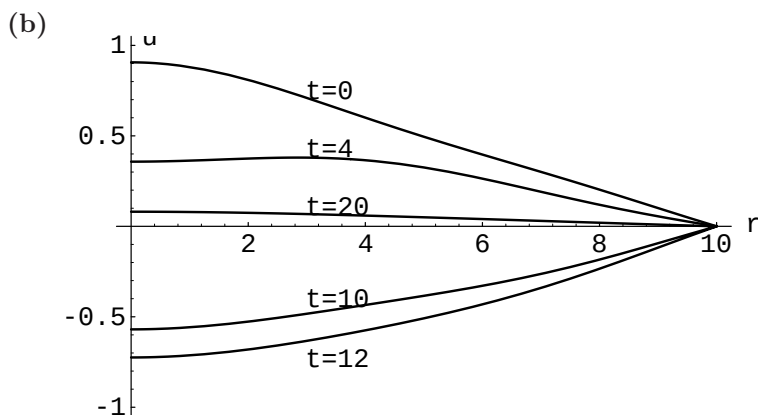
$$A_2 = \frac{2}{100J_1^2(x_2)} \int_0^{10} r J_0(x_2 r/10)(1 - r/10) dr = 0.0687,$$

and

$$A_3 = \frac{2}{100J_1^2(x_3)} \int_0^{10} r J_0(x_3 r/10)(1 - r/10) dr = 0.0531.$$

Since $g(r) = 0$, $B_n = 0$, $n = 1, 2, 3, \dots$, and the third partial sum of the series solution is

$$\begin{aligned}
 S_3(r, t) &= \sum_{n=1}^{\infty} A_n \cos(x_n t/10) J_0(x_n r/10) \\
 &= 0.7845 \cos(0.2405t) J_0(0.2405r) + 0.0687 \cos(0.5520t) J_0(0.5520r) \\
 &\quad + 0.0531 \cos(0.8654t) J_0(0.8654r).
 \end{aligned}$$



14. Because of the nonhomogeneous boundary condition $u(c, t) = 200$ we use the substitution $u(r, t) = v(r, t) + \psi(r)$. This gives

$$k \left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \psi'' + \frac{1}{r} \psi' \right) = \frac{\partial v}{\partial t}.$$

This equation will be homogeneous provided $\psi'' + (1/r)\psi' = 0$ or $\psi(r) = c_1 \ln r + c_2$. Since $\ln r$ is unbounded as $r \rightarrow 0$ we take $c_1 = 0$. Then $\psi(r) = c_2$ and using $u(c, t) = v(c, t) + c_2 = 200$ we set $c_2 = 200$, giving $v(c, t) = 0$. Referring to Problem 5 in this section, the solution of the boundary-value problem

$$k \left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} \right) = \frac{\partial v}{\partial t}, \quad 0 < r < c, \quad t > 0$$

$$\begin{aligned} v(c, t) &= 0, \quad t > 0 \\ v(r, 0) &= -200, \quad 0 < r < c \end{aligned}$$

is

$$v(r, t) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) e^{-\alpha_n^2 k t},$$

where the separation constant is $-\lambda = -\alpha^2$. The eigenvalues are $\lambda_n = \alpha^2 = x_n^2/c^2$ where x_n is a positive root of $J_0(x) = 0$ and the coefficients A_n are

$$A_n = \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^c r J_0(\alpha_n r) (-200) dr = -\frac{400}{c^2 J_1^2(\alpha_n c)} \int_0^c r J_0(\alpha_n r) dr.$$

Taking $c = 10$ and $k = 0.1$ we have

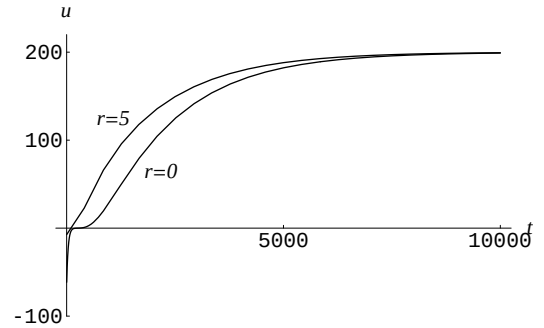
$$u(r, t) = v(r, t) + 200 = 200 + \sum_{n=1}^{\infty} A_n J_0(x_n r/10) e^{-0.01 x_n^2 t/100}$$

where

$$A_n = -\frac{4}{J_1^2(x_n)} \int_0^{10} r J_0(x_n r/10) dr.$$

Using a CAS we find that the first five values of x_n are $x_1 = 2.4048$, $x_2 = 5.5201$, $x_3 = 8.6537$, $x_4 = 11.7915$, and $x_5 = 14.9309$, with corresponding eigenvalues $\lambda_1 = 0.0578$, $\lambda_2 = 0.3047$, $\lambda_3 = 0.7489$, $\lambda_4 = 1.3904$, and $\lambda_5 = 2.2293$. The first five values of A_n are $A_1 = -320.4$, $A_2 = 213.0$, $A_3 = -170.3$, $A_4 = 145.9$, and $A_5 = -129.7$. Using a root finding application in a CAS we find that $u(5, t) = 100$ when $t \approx 1331$ and

$u(0, t) = 100$ when $t \approx 2005$. Since $u = 200$ is an asymptote for the graphs of $u(0, t)$ and $u(5, t)$ we solve $u(5, t) = 199.9$ and $u(0, t) = 199.9$. This gives $t \approx 13,265$ and $t \approx 13,958$, respectively.



15. (a) The boundary-value problem is

$$\begin{aligned} a^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} \right) &= \frac{\partial^2 u}{\partial t^2}, \quad 0 < r < 1, \quad t > 0 \\ u(1, t) &= 0, \quad t > 0 \\ u(r, 0) &= 0, \quad \frac{\partial u}{\partial t} \Big|_{t=0} = \begin{cases} -v_0, & 0 \leq r < b \\ 0, & b \leq r < 1 \end{cases}, \quad 0 < r < 1, \end{aligned}$$

and the solution is

$$u(r, t) = \sum_{n=1}^{\infty} (A_n \cos a \alpha_n t + B_n \sin a \alpha_n t) J_0(\alpha_n r),$$

where the eigenvalues $\lambda_n = \alpha_n^2$ are defined by $J_0(\alpha) = 0$ and $A_n = 0$ since $f(r) = 0$. The coefficients B_n

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are given by

$$\begin{aligned}
 B_n &= \frac{2}{a\alpha_n J_1^2(\alpha_n)} \int_0^b r J_0(\alpha_n r) g(r) dr = -\frac{2v_0}{a\alpha_n J_1^2(\alpha_n)} \int_0^b r J_0(\alpha_n r) dr \\
 &\quad \boxed{\text{let } x = \alpha_n r} \\
 &= -\frac{2v_0}{a\alpha_n J_1^2(\alpha_n)} \int_0^{\alpha_n b} \frac{x}{\alpha_n} J_0(x) \frac{1}{\alpha_n} dx = -\frac{2v_0}{a\alpha_n^3 J_1^2(\alpha_n)} \int_0^{\alpha_n b} x J_0(x) dx \\
 &= -\frac{2v_0}{a\alpha_n^3 J_1^2(\alpha_n)} (x J_1(x)) \Big|_0^{\alpha_n b} = -\frac{2v_0}{a\alpha_n^3 J_1(\alpha_n)} (\alpha_n b J_1(\alpha_n b)) = -\frac{2v_0 b}{a\alpha_n^2} \frac{J_1(\alpha_n b)}{J_1^2(\alpha_n)}.
 \end{aligned}$$

Thus,

$$u(r, t) = \frac{-2v_0 b}{a} \sum_{n=1}^{\infty} \frac{1}{\alpha_n^2} \frac{J_1(\alpha_n b)}{J_1^2(\alpha_n)} \sin(a\alpha_n t) J_0(\alpha_n r).$$

- (b) The standing wave $u_n(r, t)$ is given by $u_n(r, t) = B_n \sin(a\alpha_n t) J_0(\alpha_n r)$, which has frequency $f_n = a\alpha_n/2\pi$, where α_n is the n th positive zero of $J_0(x)$. The fundamental frequency is $f_1 = a\alpha_1/2\pi$. The next two frequencies are

$$f_2 = \frac{a\alpha_2}{2\pi} = \frac{\alpha_2}{\alpha_1} \left(\frac{a\alpha_1}{2\pi} \right) = \frac{5.520}{2.405} f_1 = 2.295 f_1$$

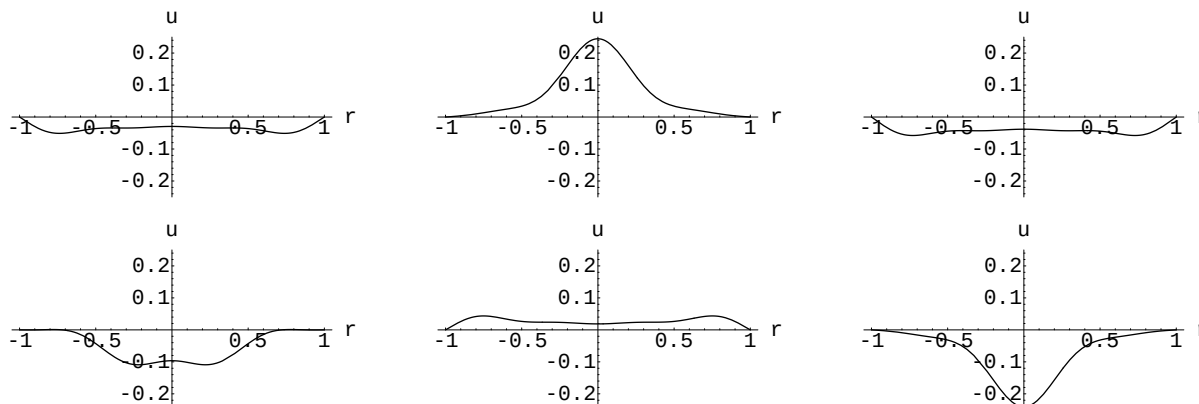
and

$$f_3 = \frac{a\alpha_3}{2\pi} = \frac{\alpha_3}{\alpha_1} \left(\frac{a\alpha_1}{2\pi} \right) = \frac{8.654}{2.405} f_1 = 3.598 f_1.$$

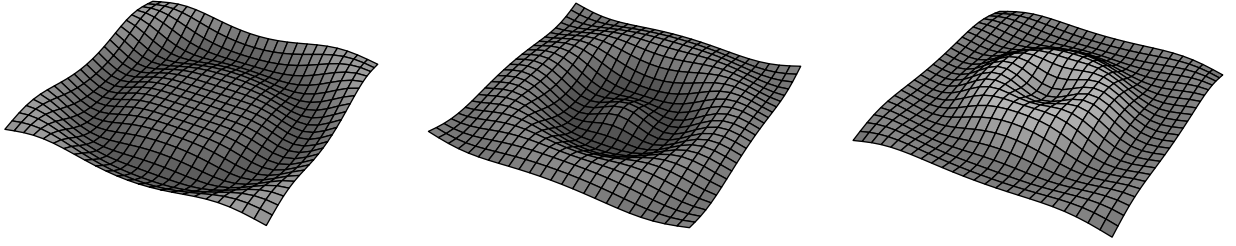
- (c) With $a = 1$, $b = \frac{1}{4}$, and $v_0 = 1$, the solution becomes

$$u(r, t) = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{\alpha_n^2} \frac{J_1(\alpha_n/4)}{J_1^2(\alpha_n)} \sin(\alpha_n t) J_0(\alpha_n r).$$

The graphs of $S_5(r, t)$ for $t = 1, 2, 3, 4, 5, 6$ are shown below.



(d) Three frames from the movie are shown.



EXERCISES 14.3

Problems in Spherical Coordinates

1. To compute

$$A_n = \frac{2n+1}{2c^n} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta$$

we substitute $x = \cos \theta$ and $dx = -\sin \theta d\theta$. Then

$$A_n = \frac{2n+1}{2c^n} \int_1^{-1} F(x) P_n(x) (-dx) = \frac{2n+1}{2c^n} \int_{-1}^1 F(x) P_n(x) dx$$

where

$$F(x) = \begin{cases} 0, & -1 < x < 0 \\ 50, & 0 < x < 1 \end{cases} = 50 \begin{cases} 0, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases}.$$

The coefficients A_n are computed in Example 3 of Section 12.6. Thus

$$\begin{aligned} u(r, \theta) &= \sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta) \\ &= 50 \left[\frac{1}{2} P_0(\cos \theta) + \frac{3}{4} \left(\frac{r}{c} \right) P_1(\cos \theta) - \frac{7}{16} \left(\frac{r}{c} \right)^3 P_3(\cos \theta) + \frac{11}{32} \left(\frac{r}{c} \right)^5 P_5(\cos \theta) + \cdots \right]. \end{aligned}$$

2. In the solution of the Cauchy-Euler equation,

$$R(r) = c_1 r^n + c_2 r^{-(n+1)},$$

we define $c_1 = 0$ since we expect the potential u to be bounded as $r \rightarrow \infty$. Hence

$$u_n(r, \theta) = A_n r^{-(n+1)} P_n(\cos \theta)$$

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^{-(n+1)} P_n(\cos \theta).$$

When $r = c$ we have

$$f(\theta) = \sum_{n=0}^{\infty} A_n c^{-(n+1)} P_n(\cos \theta)$$

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so that

$$A_n = c^{n+1} \frac{(2n+1)}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

The solution of the problem is then

$$u(r, \theta) = \sum_{n=0}^{\infty} \left(\frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta \right) \left(\frac{c}{r} \right)^{n+1} P_n(\cos \theta).$$

3. The coefficients are given by

$$\begin{aligned} A_n &= \frac{2n+1}{2c^n} \int_0^\pi \cos \theta P_n(\cos \theta) \sin \theta d\theta = \frac{2n+1}{2c^n} \int_0^\pi P_1(\cos \theta) P_n(\cos \theta) \sin \theta d\theta \\ &\quad \boxed{x = \cos \theta, \quad dx = -\sin \theta d\theta} \\ &= \frac{2n+1}{2c^n} \int_{-1}^1 P_1(x) P_n(x) dx. \end{aligned}$$

Since $P_n(x)$ and $P_m(x)$ are orthogonal for $m \neq n$, $A_n = 0$ for $n \neq 1$ and

$$A_1 = \frac{2(1)+1}{2c^1} \int_{-1}^1 P_1(x) P_1(x) dx = \frac{3}{2c} \int_{-1}^1 x^2 dx = \frac{1}{c}.$$

Thus

$$u(r, \theta) = \frac{r}{c} P_1(\cos \theta) = \frac{r}{c} \cos \theta.$$

4. The coefficients are given by

$$A_n = \frac{2n+1}{2c^n} \int_0^\pi (1 - \cos 2\theta) P_n(\cos \theta) \sin \theta d\theta.$$

These were computed in Problem 18 of Section 12.6. Thus

$$u(r, \theta) = \frac{4}{3} P_0(\cos \theta) - \frac{4}{3} \left(\frac{r}{c} \right)^2 P_2(\cos \theta).$$

5. Referring to Example 1 in the text we have

$$\Theta = P_n(\cos \theta) \quad \text{and} \quad R = c_1 r^n + c_2 r^{-(n+1)}.$$

Since $u(b, \theta) = R(b)\Theta(\theta) = 0$,

$$c_1 b^n + c_2 b^{-(n+1)} = 0 \quad \text{or} \quad c_1 = -c_2 b^{-2n-1},$$

and

$$R(r) = -c_2 b^{-2n-1} r^n + c_2 r^{-(n+1)} = c_2 \left(\frac{b^{2n+1} - r^{2n+1}}{b^{2n+1} r^{n+1}} \right).$$

Then

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n \frac{b^{2n+1} - r^{2n+1}}{b^{2n+1} r^{n+1}} P_n(\cos \theta)$$

where

$$\frac{b^{2n+1} - a^{2n+1}}{b^{2n+1} a^{n+1}} A_n = \frac{2n+1}{2} \int_0^\pi f(\theta) P_n(\cos \theta) \sin \theta d\theta.$$

6. Referring to Example 1 in the text we have

$$R(r) = c_1 r^n \quad \text{and} \quad \Theta(\theta) = P_n(\cos \theta).$$

Now $\Theta(\pi/2) = 0$ implies that n is odd, so

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{2n+1} P_{2n+1}(\cos \theta).$$

From

$$u(c, \theta) = f(\theta) = \sum_{n=0}^{\infty} A_{2n+1} c^{2n+1} P_{2n+1}(\cos \theta)$$

we see that

$$A_{2n+1} c^{2n+1} = (4n+3) \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

Thus

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{2n+1} P_{2n+1}(\cos \theta)$$

where

$$A_{2n+1} = \frac{4n+3}{c^{2n+1}} \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

7. Referring to Example 1 in the text we have

$$\begin{aligned} r^2 R'' + 2r R' - \lambda R &= 0 \\ \sin \theta \Theta'' + \cos \theta \Theta' + \lambda \sin \theta \Theta &= 0. \end{aligned}$$

Substituting $x = \cos \theta$, $0 \leq \theta \leq \pi/2$, the latter equation becomes

$$(1-x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \lambda \Theta = 0, \quad 0 \leq x \leq 1.$$

Taking the solutions of this equation to be the Legendre polynomials $P_n(x)$ corresponding to $\lambda = n(n+1)$ for $n = 1, 2, 3, \dots$, we have $\Theta = P_n(\cos \theta)$. Since

$$\left. \frac{\partial u}{\partial \theta} \right|_{\theta=\pi/2} = \Theta'(\pi/2) R(r) = 0$$

we have

$$\Theta'(\pi/2) = -(\sin \pi/2) P'_n(\cos \pi/2) = -P'_n(0) = 0.$$

As noted in the hint, $P'_n(0) = 0$ only if n is even. Thus $\Theta = P_n(\cos \theta)$, $n = 0, 2, 4, \dots$. As in Example 1, $R(r) = c_1 r^n$. Hence

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n} r^{2n} P_{2n}(\cos \theta).$$

At $r = c$,

$$f(\theta) = \sum_{n=0}^{\infty} A_{2n} c^{2n} P_{2n}(\cos \theta).$$

Using Problem 19 in Section 12.6, we obtain

$$c^{2n} A_{2n} = (4n+1) \int_{\pi/2}^0 f(\theta) P_{2n}(\cos \theta) (-\sin \theta) d\theta$$

and

$$A_{2n} = \frac{4n+1}{c^{2n}} \int_0^{\pi/2} f(\theta) \sin \theta P_{2n}(\cos \theta) d\theta.$$

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8. Referring to Example 1 in the text we have

$$R(r) = c_1 r^n + c_2 r^{-(n+1)} \quad \text{and} \quad \Theta(\theta) = P_n(\cos \theta).$$

Since we expect $u(r, \theta)$ to be bounded as $r \rightarrow \infty$, we define $c_1 = 0$. Also $\Theta(\pi/2) = 0$ implies that n is odd, so

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{-2(n+1)} P_{2n+1}(\cos \theta).$$

From

$$u(c, \theta) = f(\theta) = \sum_{n=0}^{\infty} A_{2n+1} c^{-2(n+1)} P_{2n+1}(\cos \theta)$$

we see that

$$A_{2n+1} c^{-2(n+1)} = (4n+3) \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

Thus

$$u(r, \theta) = \sum_{n=0}^{\infty} A_{2n+1} r^{-2(n+1)} P_{2n+1}(\cos \theta)$$

where

$$A_{2n+1} = (4n+3) c^{2(n+1)} \int_0^{\pi/2} f(\theta) \sin \theta P_{2n+1}(\cos \theta) d\theta.$$

9. Checking the hint, we find

$$\frac{1}{r} \frac{\partial^2}{\partial r^2}(ru) = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{\partial u}{\partial r} + u \right] = \frac{1}{r} \left[r \frac{\partial^2 u}{\partial r^2} + \frac{\partial u}{\partial r} + \frac{\partial u}{\partial r} \right] = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r}.$$

The partial differential equation then becomes

$$\frac{\partial^2}{\partial r^2}(ru) = r \frac{\partial u}{\partial t}.$$

Now, letting $ru(r, t) = v(r, t) + \psi(r)$, since the boundary condition is nonhomogeneous, we obtain

$$\frac{\partial^2}{\partial r^2} [v(r, t) + \psi(r)] = r \frac{\partial}{\partial t} \left[\frac{1}{r} v(r, t) + \psi(r) \right]$$

or

$$\frac{\partial^2 v}{\partial r^2} + \psi''(r) = \frac{\partial v}{\partial t}.$$

This differential equation will be homogeneous if $\psi''(r) = 0$ or $\psi(r) = c_1 r + c_2$. Now

$$u(r, t) = \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) \quad \text{and} \quad \frac{1}{r} \psi(r) = c_1 + \frac{c_2}{r}.$$

Since we want $u(r, t)$ to be bounded as r approaches 0, we require $c_2 = 0$. Then $\psi(r) = c_1 r$. When $r = 1$

$$u(1, t) = v(1, t) + \psi(1) = v(1, t) + c_1 = 100,$$

and we will have the homogeneous boundary condition $v(1, t) = 0$ when $c_1 = 100$. Consequently, $\psi(r) = 100r$.

The initial condition

$$u(r, 0) = \frac{1}{r} v(r, 0) + \frac{1}{r} \psi(r) = \frac{1}{r} v(r, 0) + 100 = 0$$

implies $v(r, 0) = -100r$. We are thus led to solve the new boundary-value problem

$$\frac{\partial^2 v}{\partial r^2} = \frac{\partial v}{\partial t}, \quad 0 < r < 1, \quad t > 0,$$

$$v(1, t) = 0, \quad \lim_{r \rightarrow 0} \frac{1}{r} v(r, t) < \infty,$$

$$v(r, 0) = -100r.$$

Letting $v(r, t) = R(r)T(t)$ and using the separation constant $-\lambda$ we obtain

$$R'' + \lambda R = 0 \quad \text{and} \quad T' + \lambda T = 0.$$

Using $\lambda = \alpha^2 > 0$ we then have

$$R(r) = c_3 \cos \alpha r + c_4 \sin \alpha r \quad \text{and} \quad T(t) = c_5 e^{-\alpha^2 t}.$$

The boundary conditions are equivalent to $R(1) = 0$ and $\lim_{r \rightarrow 0} R(r)/r < \infty$. Since

$$\lim_{r \rightarrow 0} \frac{\cos \alpha r}{r}$$

does not exist we must have $c_3 = 0$. Then $R(r) = c_4 \sin \alpha r$, and $R(1) = 0$ implies $\alpha = n\pi$ for $n = 1, 2, 3, \dots$.

Thus

$$v_n(r, t) = A_n e^{-n^2 \pi^2 t} \sin n\pi r$$

for $n = 1, 2, 3, \dots$. Using the condition $\lim_{r \rightarrow 0} R(r)/r < \infty$ it is easily shown that there are no eigenvalues for $\lambda = 0$, nor does setting the common constant to $-\lambda = \alpha^2$ when separating variables lead to any solutions. Now, by the superposition principle,

$$v(r, t) = \sum_{n=1}^{\infty} A_n e^{-n^2 \pi^2 t} \sin n\pi r.$$

The initial condition $v(r, 0) = -100r$ implies

$$-100r = \sum_{n=1}^{\infty} A_n \sin n\pi r.$$

This is a Fourier sine series and so

$$\begin{aligned} A_n &= 2 \int_0^1 (-100r \sin n\pi r) dr = -200 \left[-\frac{r}{n\pi} \cos n\pi r \Big|_0^1 + \int_0^1 \frac{1}{n\pi} \cos n\pi r dr \right] \\ &= -200 \left[-\frac{\cos n\pi}{n\pi} + \frac{1}{n^2 \pi^2} \sin n\pi r \Big|_0^1 \right] = -200 \left[-\frac{(-1)^n}{n\pi} \right] = \frac{(-1)^n 200}{n\pi}. \end{aligned}$$

A solution of the problem is thus

$$\begin{aligned} u(r, t) &= \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) = \frac{1}{r} \sum_{n=1}^{\infty} (-1)^n \frac{20}{n\pi} e^{-n^2 \pi^2 t} \sin n\pi r + \frac{1}{r} (100r) \\ &= \frac{200}{\pi r} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} e^{-n^2 \pi^2 t} \sin n\pi r + 100. \end{aligned}$$

10. Referring to Problem 9 we have

$$\frac{\partial^2 v}{\partial r^2} + \psi''(r) = \frac{\partial v}{\partial t}$$

where $\psi(r) = c_1 r$. Since

$$u(r, t) = \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) = \frac{1}{r} v(r, t) + c_1$$

we have

$$\frac{\partial u}{\partial r} = \frac{1}{r} v_r(r, t) - \frac{1}{r^2} v(r, t).$$

When $r = 1$,

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} = v_r(1, t) - v(1, t)$$

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and

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} + hu(1, t) = v_r(1, t) - v(1, t) + h[v(1, t) + \psi(1)] = v_r(1, t) + (h-1)v(1, t) + hc_1.$$

Thus the boundary condition

$$\left. \frac{\partial u}{\partial r} \right|_{r=1} + hu(1, t) = hu_1$$

will be homogeneous when $hc_1 = hu_1$ or $c_1 = u_1$. Consequently $\psi(r) = u_1 r$. The initial condition

$$u(r, 0) = \frac{1}{r}v(r, 0) + \frac{1}{r}\psi(r) = \frac{1}{r}v(r, 0) + u_1 = u_0$$

implies $v(r, 0) = (u_0 - u_1)r$. We are thus led to solve the new boundary-value problem

$$\begin{aligned} \frac{\partial^2 v}{\partial r^2} &= \frac{\partial v}{\partial t}, & 0 < r < 1, & \quad t > 0, \\ v_r(1, t) + (h-1)v(1, t) &= 0, & t > 0, \\ \lim_{r \rightarrow 0} \frac{1}{r}v(r, t) &< \infty, \\ v(r, 0) &= (u_0 - u_1)r. \end{aligned}$$

Separating variables as in Problem 9 leads to

$$R(r) = c_3 \cos \alpha r + c_4 \sin \alpha r \quad \text{and} \quad T(t) = c_5 e^{-\alpha^2 t}.$$

The boundary conditions are equivalent to

$$R'(1) + (h-1)R(1) = 0 \quad \text{and} \quad \lim_{r \rightarrow 0} \frac{1}{r}R(r) < \infty.$$

As in Problem 6 we use the second condition to determine that $c_3 = 0$ and $R(r) = c_4 \sin \alpha r$. Then

$$R'(1) + (h-1)R(1) = c_4 \alpha \cos \alpha + c_4(h-1) \sin \alpha = 0$$

and the α_n are the consecutive nonnegative roots of $\tan \alpha = \alpha/(1-h)$. Now

$$v(r, t) = \sum_{n=1}^{\infty} A_n e^{-\alpha_n^2 t} \sin \alpha_n r.$$

From

$$v(r, 0) = (u_0 - u_1)r = \sum_{n=1}^{\infty} A_n \sin \alpha_n r$$

we obtain

$$A_n = \frac{\int_0^1 (u_0 - u_1)r \sin \alpha_n r \, dr}{\int_0^1 \sin^2 \alpha_n r \, dr}.$$

We compute the integrals

$$\int_0^1 r \sin \alpha_n r \, dr = \left(\frac{1}{\alpha_n^2} \sin \alpha_n r - \frac{1}{\alpha_n} \cos \alpha_n r \right) \Big|_0^1 = \frac{1}{\alpha_n^2} \sin \alpha_n - \frac{1}{\alpha_n} \cos \alpha_n$$

and

$$\int_0^1 \sin^2 \alpha_n r \, dr = \left(\frac{1}{2}r - \frac{1}{4\alpha_n} \sin 2\alpha_n r \right) \Big|_0^1 = \frac{1}{2} - \frac{1}{4\alpha_n} \sin 2\alpha_n.$$

Using $\alpha_n \cos \alpha_n = -(h-1) \sin \alpha_n$ we then have

$$\begin{aligned} A_n &= (u_0 - u_1) \frac{\frac{1}{\alpha_n^2} \sin \alpha_n - \frac{1}{\alpha_n} \cos \alpha_n}{\frac{1}{2} - \frac{1}{4\alpha_n} \sin 2\alpha_n} = (u_0 - u_1) \frac{4 \sin \alpha_n - 4\alpha_n \cos \alpha_n}{2\alpha_n^2 - \alpha_n \sin 2\alpha_n} \\ &= 2(u_0 - u_1) \frac{\sin \alpha_n + (h-1) \sin \alpha_n}{\alpha_n^2 + (h-1) \sin \alpha_n \sin \alpha_n} = 2(u_0 - u_1) h \frac{\sin \alpha_n}{\alpha_n^2 + (h-1) \sin^2 \alpha_n}. \end{aligned}$$

Therefore

$$u(r, t) = \frac{1}{r} v(r, t) + \frac{1}{r} \psi(r) = u_1 + 2(u_0 - u_1) h \sum_{n=1}^{\infty} \frac{\sin \alpha_n \sin \alpha_n r}{r[\alpha_n^2 + (h-1) \sin^2 \alpha_n]} e^{-\alpha_n^2 t}.$$

11. We write the differential equation in the form

$$a^2 \frac{1}{r} \frac{\partial^2}{\partial r^2} (ru) = \frac{\partial^2 u}{\partial t^2} \quad \text{or} \quad a^2 \frac{\partial^2}{\partial r^2} (ru) = r \frac{\partial^2 u}{\partial t^2},$$

and then let $v(r, t) = ru(r, t)$. The new boundary-value problem is

$$\begin{aligned} a^2 \frac{\partial^2 v}{\partial r^2} &= \frac{\partial^2 v}{\partial t^2}, \quad 0 < r < c, \quad t > 0 \\ v(c, t) &= 0, \quad t > 0 \\ v(r, 0) &= rf(r), \quad \left. \frac{\partial v}{\partial t} \right|_{t=0} = rg(r). \end{aligned}$$

Letting $v(r, t) = R(r)T(t)$ and using the separation constant $-\lambda = -\alpha^2$ we obtain

$$\begin{aligned} R'' + \alpha^2 R &= 0 \\ T'' + a^2 \alpha^2 T &= 0 \end{aligned}$$

and

$$\begin{aligned} R(r) &= c_1 \cos \alpha r + c_2 \sin \alpha r \\ T(t) &= c_3 \cos a\alpha t + c_4 \sin a\alpha t. \end{aligned}$$

Since $u(r, t) = v(r, t)/r$, in order to insure boundedness at $r = 0$ we define $c_1 = 0$. Then $R(r) = c_2 \sin \alpha r$ and the condition $R(c) = 0$ implies $\alpha = n\pi/c$. Thus

$$v(r, t) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{c} t + B_n \sin \frac{n\pi a}{c} t \right) \sin \frac{n\pi}{c} r.$$

From

$$v(r, 0) = rf(r) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{c} r$$

we see that

$$A_n = \frac{2}{c} \int_0^c rf(r) \sin \frac{n\pi}{c} r dr.$$

From

$$\left. \frac{\partial v}{\partial t} \right|_{t=0} = rg(r) = \sum_{n=1}^{\infty} \left(B_n \frac{n\pi a}{c} \right) \sin \frac{n\pi}{c} r$$

we see that

$$B_n = \frac{c}{n\pi a} \cdot \frac{2}{c} \int_0^c rg(r) \sin \frac{n\pi}{c} r dr = \frac{2}{n\pi a} \int_0^c rg(r) \sin \frac{n\pi}{c} r dr.$$

The solution is

$$u(r, t) = \frac{1}{r} \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi a}{c} t + B_n \sin \frac{n\pi a}{c} t \right) \sin \frac{n\pi}{c} r,$$

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where A_n and B_n are given above.

12. Proceeding as in Example 1 we obtain

$$\Theta(\theta) = P_n(\cos \theta) \quad \text{and} \quad R(r) = c_1 r^n + c_2 r^{-(n+1)}$$

so that

$$u(r, \theta) = \sum_{n=0}^{\infty} (A_n r^n + B_n r^{-(n+1)}) P_n(\cos \theta).$$

To satisfy $\lim_{r \rightarrow \infty} u(r, \theta) = -Er \cos \theta$ we must have $A_n = 0$ for $n = 2, 3, 4, \dots$. Then

$$\lim_{r \rightarrow \infty} u(r, \theta) = -Er \cos \theta = A_0 \cdot 1 + A_1 r \cos \theta,$$

so $A_0 = 0$ and $A_1 = -E$. Thus

$$u(r, \theta) = -Er \cos \theta + \sum_{n=0}^{\infty} B_n r^{-(n+1)} P_n(\cos \theta).$$

Now

$$u(c, \theta) = 0 = -Ec \cos \theta + \sum_{n=0}^{\infty} B_n c^{-(n+1)} P_n(\cos \theta)$$

so

$$\sum_{n=0}^{\infty} B_n c^{-(n+1)} P_n(\cos \theta) = Ec \cos \theta$$

and

$$B_n c^{-(n+1)} = \frac{2n+1}{2} \int_0^\pi Ec \cos \theta P_n(\cos \theta) \sin \theta d\theta.$$

Now $\cos \theta = P_1(\cos \theta)$ so, for $n \neq 1$,

$$\int_0^\pi \cos \theta P_n(\cos \theta) \sin \theta d\theta = 0$$

by orthogonality. Thus $B_n = 0$ for $n \neq 1$ and

$$B_1 = \frac{3}{2} Ec^3 \int_0^\pi \cos^2 \theta \sin \theta d\theta = Ec^3.$$

Therefore,

$$u(r, \theta) = -Er \cos \theta + Ec^3 r^{-2} \cos \theta.$$

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1. We have

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^\pi u_0 \, d\theta + \frac{1}{2\pi} \int_\pi^{2\pi} (-u_0) \, d\theta = 0 \\ A_n &= \frac{1}{c^n \pi} \int_0^\pi u_0 \cos n\theta \, d\theta + \frac{1}{c^n \pi} \int_\pi^{2\pi} (-u_0) \cos n\theta \, d\theta = 0 \\ B_n &= \frac{1}{c^n \pi} \int_0^\pi u_0 \sin n\theta \, d\theta + \frac{1}{c^n \pi} \int_\pi^{2\pi} (-u_0) \sin n\theta \, d\theta = \frac{2u_0}{c^n n \pi} [1 - (-1)^n] \end{aligned}$$

and so

$$u(r, \theta) = \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \left(\frac{r}{c}\right)^n \sin n\theta.$$

2. We have

$$\begin{aligned} A_0 &= \frac{1}{2\pi} \int_0^{\pi/2} d\theta + \frac{1}{2\pi} \int_{3\pi/2}^{2\pi} d\theta = \frac{1}{2} \\ A_n &= \frac{1}{c^n \pi} \int_0^{\pi/2} \cos n\theta \, d\theta + \frac{1}{c^n \pi} \int_{3\pi/2}^{2\pi} \cos n\theta \, d\theta = \frac{1}{c^n n \pi} \left[\sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2} \right] \\ B_n &= \frac{1}{c^n \pi} \int_0^{\pi/2} \sin n\theta \, d\theta + \frac{1}{c^n \pi} \int_{3\pi/2}^{2\pi} \sin n\theta \, d\theta = \frac{1}{c^n n \pi} \left[\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2} \right] \end{aligned}$$

and so

$$u(r, \theta) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \left(\frac{r}{c}\right)^n \left[\frac{\sin \frac{n\pi}{2} - \sin \frac{3n\pi}{2}}{n} \cos n\theta + \frac{\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2}}{n} \sin n\theta \right].$$

3. The conditions $\Theta(0) = 0$ and $\Theta(\pi) = 0$ applied to $\Theta = c_1 \cos \alpha\theta + c_2 \sin \alpha\theta$ give $c_1 = 0$ and $\alpha = n$, $n = 1, 2, 3, \dots$, respectively. Thus we have the Fourier sine-series coefficients

$$A_n = \frac{2}{\pi} \int_0^\pi u_0(\pi\theta - \theta^2) \sin n\theta \, d\theta = \frac{4u_0}{n^3 \pi} [1 - (-1)^n].$$

Thus

$$u(r, \theta) = \frac{4u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} r^n \sin n\theta.$$

4. In this case

$$A_n = \frac{2}{\pi} \int_0^\pi \sin \theta \sin n\theta \, d\theta = \frac{1}{\pi} \int_0^\pi [\cos(1-n)\theta - \cos(1+n)\theta] \, d\theta = 0, \quad n \neq 1.$$

For $n = 1$,

$$A_1 = \frac{2}{\pi} \int_0^\pi \sin^2 \theta \, d\theta = \frac{1}{\pi} \int_0^\pi (1 - \cos 2\theta) \, d\theta = 1.$$

Thus

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^n \sin n\theta$$

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reduces to

$$u(r, \theta) = r \sin \theta.$$

5. The insulation conditions are $u_\theta(r, 0) = 0$ and $u_\theta(r, \pi) = 0$. The eigenvalue $\lambda = 0$ corresponds to a constant eigenfunction. The other eigenvalues are $\lambda = n^2$, $n = 1, 2, 3, \dots$, with corresponding eigenfunctions $\Theta(\theta) = c_1 \cos n\theta$. Also, assuming a bounded solution at $r = 0$, $R(r) = c_3 r^n$ and so

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} A_n r^n \cos n\theta.$$

At $r = c$

$$f(\theta) = A_0 + \sum_{n=1}^{\infty} A_n c^n \cos n\theta,$$

from which we see that

$$A_0 = \frac{a_0}{2} = \frac{1}{\pi} \int_0^\pi f(\theta) d\theta \quad \text{and} \quad A_n = \frac{2}{\pi} \int_0^\pi f(\theta) \cos n\theta d\theta.$$

6. Two of the boundary conditions are $u(r, 0) = 0$ and $u_\theta(r, \pi) = 0$ which imply $\Theta(0) = 0$ and $\Theta'(\pi) = 0$. For $\lambda = [(2n-1)/2]^2$, $n = 1, 2, 3, \dots$, we have $\Theta(\theta) = c_2 \sin[(2n-1)/2]\theta$. Assuming a bounded solution at $r = 0$ we have $R(r) = c_3 r^n$, so

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^n \sin\left(\frac{2n-1}{2}\theta\right) \quad \text{and} \quad f(\theta) = \sum_{n=1}^{\infty} A_n c^n \sin\left(\frac{2n-1}{2}\theta\right).$$

This is not a Fourier series but it is an orthogonal series expansion of f , so

$$A_n = c^{-n} \left(\frac{\int_0^\pi f(\theta) \sin\left(\frac{2n-1}{2}\theta\right) d\theta}{\int_0^\pi \sin^2\left(\frac{2n-1}{2}\theta\right) d\theta} \right).$$

7. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad 0 < \theta < \frac{\pi}{4}, \quad \frac{1}{2} < r < 1,$$

$$u(r, 0) = 0, \quad u(r, \pi/4) = 0, \quad \frac{1}{2} < r < 1,$$

$$u(1/2, \theta) = u_0, \quad u_r(1, \theta) = 0, \quad 0 < \theta < \frac{\pi}{4}.$$

Proceeding as in Example 1 in Section 14.1 using the separation constant $\lambda = \alpha^2$ we obtain

$$r^2 R'' + rR' - \lambda R = 0$$

$$\Theta'' + \lambda \Theta = 0$$

with solutions

$$\Theta(\theta) = c_1 \cos \alpha \theta + c_2 \sin \alpha \theta$$

$$R(r) = c_3 r^\alpha + c_4 r^{-\alpha}.$$

Applying the boundary conditions $\Theta(0) = 0$ and $\Theta(\pi/4) = 0$ gives $c_1 = 0$ and $\alpha = 4n$ for $n = 1, 2, 3, \dots$. From $R_r(1) = 0$ we obtain $c_3 = c_4$. Therefore

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n (r^{4n} + r^{-4n}) \sin 4n\theta.$$

From

$$u(1/2, \theta) = u_0 = \sum_{n=1}^{\infty} A_n \left(\frac{1}{2^{4n}} + \frac{1}{2^{-4n}} \right) \sin 4n\theta$$

we find

$$A_n \left(\frac{1}{2^{4n}} + \frac{1}{2^{-4n}} \right) = \frac{2}{\pi/4} \int_0^{\pi/4} u_0 \sin 4n\theta \, d\theta = \frac{2u_0}{n\pi} [1 - (-1)^n]$$

or

$$A_n = \frac{2u_0}{n\pi(2^{4n} + 2^{-4n})} [1 - (-1)^n].$$

Thus the steady-state temperature in the plate is

$$u(r, \theta) = \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{[r^{4n} + r^{-4n}][1 - (-1)^n]}{n[2^{4n} + 2^{-4n}]} \sin 4n\theta.$$

8. The boundary-value problem in polar coordinates is

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad 0 < \theta < \beta, \quad a < r < b, \\ u(a, \theta) &= 0, \quad u(b, \theta) = f(\theta), \quad 0 < \theta < \beta, \\ u(r, 0) &= u_0, \quad u(r, \beta) = u_1, \quad a < r < b. \end{aligned}$$

Since the boundary conditions are nonhomogeneous at $\theta = 0$ and $\theta = \beta$ we let $u(r, \theta) = v(r, \theta) + \psi(\theta)$. Substituting into the partial differential equation we obtain

$$v_{rr} + \frac{1}{r} v_r + \frac{1}{r^2} (v_{\theta\theta} + \psi'') = 0.$$

This equation will be homogeneous if we require $\psi'' = 0$. In this case $\psi(\theta) = c_1 + c_2\theta$. The boundary conditions $\psi(0) = u_0$ and $\psi(\beta) = u_1$ then give $c_1 = u_0$ and $c_2 = (u_1 - u_0)/\beta$. Hence

$$\psi(\theta) = u_0 + \frac{u_1 - u_0}{\beta} \theta.$$

The homogeneous boundary-value problem for $v(r, \theta)$ is then

$$\begin{aligned} \frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} &= 0, \quad 0 < \theta < \beta, \quad a < r < b, \\ v(a, \theta) &= -u_0 - \frac{u_1 - u_0}{\beta} \theta, \quad 0 < \theta < \beta, \\ v(b, \theta) &= f(\theta) - u_0 - \frac{u_1 - u_0}{\beta} \theta, \quad 0 < \theta < \beta, \\ v(r, 0) &= 0, \quad v(r, \beta) = 0, \quad a < r < b. \end{aligned}$$

Letting $v(r, \theta) = R(r)\Theta(\theta)$ and separating variables gives

$$\Theta(\theta) = c_3 \cos \alpha\theta + c_4 \sin \alpha\theta \quad \text{and} \quad R(r) = c_5 r^\alpha + c_6 r^{-\alpha}.$$

The boundary conditions $\Theta(0) = 0$ and $\Theta(\beta) = 0$ give $c_3 = 0$ and $\alpha = n\pi/\beta$. Thus, $\Theta(\theta) = c_4 \sin n\pi\theta/\beta$ and

$$v(r, \theta) = \sum_{n=1}^{\infty} (A_n r^{n\pi/\beta} + B_n r^{-n\pi/\beta}) \sin \frac{n\pi}{\beta} \theta.$$

At $r = a$ we have

$$-u_0 - \frac{u_1 - u_0}{\beta} \theta = \sum_{n=1}^{\infty} (A_n a^{n\pi/\beta} + B_n a^{-n\pi/\beta}) \sin \frac{n\pi}{\beta} \theta,$$

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so

$$A_n a^{n\pi/\beta} + B_n a^{-n\pi/\beta} = -\frac{2}{\beta} \int_0^\beta \left(u_0 + \frac{u_1 - u_0}{\beta} \theta \right) \sin \frac{n\pi}{\beta} \theta d\theta.$$

Similarly, at $r = b$ we have

$$A_n b^{n\pi/\beta} + B_n b^{-n\pi/\beta} = \frac{2}{\beta} \int_0^\beta \left(f(\theta) - u_0 - \frac{u_1 - u_0}{\beta} \theta \right) \sin \frac{n\pi}{\beta} \theta d\theta.$$

Solving the above two simultaneous equations for A_n and B_n we get

$$A_n = \frac{-\frac{2}{\beta b^{n\pi/\beta}} \int_0^\beta \left(u_0 + \frac{u_1 - u_0}{\beta} \theta \right) \sin \frac{n\pi}{\beta} \theta d\theta - \frac{2}{\beta a^{n\pi/\beta}} \int_0^\beta \left(f(\theta) - u_0 - \frac{u_1 - u_0}{\beta} \theta \right) \sin \frac{n\pi}{\beta} \theta d\theta}{\left(\frac{a}{b}\right)^{n\pi/\beta} - \left(\frac{b}{a}\right)^{n\pi/\beta}}$$

and

$$B_n = \frac{-\frac{2a^{n\pi/\beta}}{\beta} \int_0^\beta \left(f(\theta) - u_0 - \frac{u_1 - u_0}{\beta} \theta \right) \sin \frac{n\pi}{\beta} \theta d\theta + \frac{2b^{n\pi/\beta}}{\beta} \int_0^\beta \left(u_0 + \frac{u_1 - u_0}{\beta} \theta \right) \sin \frac{n\pi}{\beta} \theta d\theta}{\left(\frac{a}{b}\right)^{n\pi/\beta} - \left(\frac{b}{a}\right)^{n\pi/\beta}}.$$

9. The boundary-value problem in polar coordinates is

$$\begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} &= 0, \quad 0 < \theta < 2\pi, \quad 1 < r < 2, \\ u(1, \theta) &= \sin^2 \theta, \quad \frac{\partial u}{\partial r} \Big|_{r=2} = 0, \quad 0 < \theta < 2\pi. \end{aligned}$$

Letting $u(r, \theta) = R(r)\Theta(\theta)$ and separating variables we obtain

$$r^2 R'' + rR' - \lambda R = 0 \quad \text{and} \quad \Theta'' + \lambda \Theta = 0.$$

For $\lambda = 0$ we have $\Theta'' = 0$ and $r^2 R'' + rR' = 0$. This gives $\Theta = c_1 + c_2 \theta$ and $R = c_3 + c_4 \ln r$. The periodicity assumption (as mentioned in Example 1 of Section 14.1 in the text) implies $c_2 = 0$, while the boundary condition $R'(2) = 0$ implies $c_4 = 0$. Thus, for $\lambda = 0$, $u = c_1 c_3 = A_0$. Now, for $\lambda = \alpha^2$, the differential equations become $\Theta'' + \alpha^2 \Theta = 0$ and $r^2 R'' + rR' - \alpha^2 R = 0$. The corresponding solutions are $\Theta = c_5 \cos \alpha \theta + c_6 \sin \alpha \theta$ and $R = c_7 r^\alpha + c_8 r^{-\alpha}$. In this case the periodicity assumption implies $\alpha = n$, $n = 1, 2, 3, \dots$, while the boundary condition $R'(2) = 0$ implies $R = c_7(r^n + 4^n r^{-n})$. The product of the solutions is $u_n = (r^n + 4^n r^{-n})(A_n \cos n\theta + B_n \sin n\theta)$ and the superposition principle implies

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} (r^n + 4^n r^{-n})(A_n \cos n\theta + B_n \sin n\theta).$$

Using the boundary condition at $r = 1$ we have

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) = A_0 + \sum_{n=1}^{\infty} (1 + 4^n)(A_n \cos n\theta + B_n \sin n\theta).$$

From this we conclude that $B_n = 0$ for all integers n , $A_0 = \frac{1}{2}$, $A_1 = 0$, $A_2 = -\frac{1}{34}$, and $A_m = 0$ for $m = 3, 4, 5, \dots$. Therefore

$$u(r, \theta) = A_0 + A_2 \cos 2\theta = \frac{1}{2} - \frac{1}{34} (r^2 + 16r^{-2}) \cos 2\theta = \frac{1}{2} - \left(\frac{1}{34} r^2 + \frac{8}{17} r^{-2} \right) \cos 2\theta.$$

10. We solve

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad r > 1, \quad 0 < \theta < \pi,$$

$$u(r, 0) = 0, \quad u(r, \pi) = 0, \quad r > 1,$$

$$u(1, \theta) = f(\theta), \quad 0 < \theta < \pi.$$

Separating variables we obtain

$$\Theta(\theta) = c_1 \cos \alpha \theta + c_2 \sin \alpha \theta$$

$$R(r) = c_3 r^\alpha + c_4 r^{-\alpha}.$$

Applying the boundary conditions $\Theta(0) = 0$, and $\Theta(\pi) = 0$ gives $c_1 = 0$ and $\alpha = n$ for $n = 1, 2, 3, \dots$. Assuming $f(\theta)$ to be bounded, we expect the solution $u(r, \theta)$ to also be bounded as $r \rightarrow \infty$. This requires that $c_3 = 0$. Therefore

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{-n} \sin n\theta.$$

From

$$u(1, \theta) = f(\theta) = \sum_{n=1}^{\infty} A_n \sin n\theta$$

we obtain

$$A_n = \frac{2}{\pi} \int_0^\pi f(\theta) \sin n\theta \, d\theta.$$

11. Letting $u(r, t) = R(r)T(t)$ and separating variables we obtain

$$\frac{R'' + \frac{1}{r}R' - hR}{R} = \frac{T'}{T} = \lambda$$

so

$$R'' + \frac{1}{r}R' - (\lambda + h)R = 0 \quad \text{and} \quad T' - \lambda T = 0.$$

From the second equation we find $T(t) = c_1 e^{\lambda t}$. If $\lambda > 0$, $T(t)$ increases without bound as $t \rightarrow \infty$. Thus we assume $\lambda = -\alpha^2 < 0$. Since $h > 0$ we can take $\mu = -\alpha^2 - h$. Then

$$R'' + \frac{1}{r}R' + \alpha^2 R = 0$$

is a parametric Bessel equation with solution

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r).$$

Since Y_0 is unbounded as $r \rightarrow 0$ we take $c_2 = 0$. Then $R(r) = c_1 J_0(\alpha r)$ and the boundary condition $u(1, t) = R(1)T(t) = 0$ implies $J_0(\alpha) = 0$. This latter equation defines the positive eigenvalues λ_n . Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) e^{(-\alpha_n^2 - h)t}.$$

From

$$u(r, 0) = 1 = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)$$

we find

$$\begin{aligned} A_n &= \frac{2}{J_1^2(\alpha_n)} \int_0^1 r J_0(\alpha_n r) \, dr & \boxed{x = \alpha_n r, \, dx = \alpha_n \, dr} \\ &= \frac{2}{J_1^2(\alpha_n)} \int_0^{\alpha_n} \frac{1}{\alpha_n^2} x J_0(x) \, dx. \end{aligned}$$

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From recurrence relation (5) in Section 12.6 of the text we have

$$xJ_0(x) = \frac{d}{dx}[xJ_1(x)].$$

Then

$$A_n = \frac{2}{\alpha_n^2 J_1^2(\alpha_n)} \int_0^{\alpha_n} \frac{d}{dx}[xJ_1(x)] dx = \frac{2}{\alpha_n^2 J_1^2(\alpha_n)} \left(xJ_1(x) \right) \Big|_0^{\alpha_n} = \frac{2\alpha_1 J_1(\alpha_n)}{\alpha_n^2 J_1^2(\alpha_n)} = \frac{2}{\alpha_n J_1(\alpha_n)}$$

and

$$u(r, t) = 2e^{-ht} \sum_{n=1}^{\infty} \frac{J_0(\alpha_n r)}{\alpha_n J_1(\alpha_n)} e^{-\alpha_n^2 t}$$

12. Letting $\lambda = \alpha^2 > 0$ and proceeding in the usual manner we find

$$u(r, t) = \sum_{n=1}^{\infty} A_n \cos a\alpha_n t J_0(\alpha_n r)$$

where the eigenvalues $\lambda_n = \alpha_n^2$ are determined by $J_0(\alpha) = 0$. Then the initial condition gives

$$u_0 J_0(x_k r) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r)$$

and so

$$A_n = \frac{2}{J_1^2(\alpha_n)} \int_0^1 r (u_0 J_0(x_k r)) J_0(\alpha_n r) dr.$$

But $J_0(\alpha) = 0$ implies that the eigenvalues are the positive zeros of J_0 , that is, $\alpha_n = x_n$ for $n = 1, 2, 3, \dots$. Therefore

$$A_n = \frac{2u_0}{J_1^2(\alpha_n)} \int_0^1 r J_0(\alpha_k r) J_0(\alpha_n r) dr = 0, \quad n \neq k$$

by orthogonality. For $n = k$,

$$A_k = \frac{2u_0}{J_1^2(\alpha_k)} \int_0^1 r J_0^2(\alpha_k r) dr = u_0$$

by (7) of Section 12.6. Thus the solution $u(r, t)$ reduces to one term when $n = k$, and

$$u(r, t) = u_0 \cos a\alpha_k t J_0(\alpha_k r) = u_0 \cos ax_k t J_0(x_k r).$$

13. Letting the separation constant be $\lambda = \alpha^2$ and referring to Example 2 in Section 14.2 in the text we have

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)$$

$$Z(z) = c_3 \cosh \alpha z + c_4 \sinh \alpha z$$

where $c_2 = 0$ and the positive eigenvalues λ_n are determined by $J_0(2\alpha) = 0$. From $Z'(0) = 0$ we obtain $c_4 = 0$. Then

$$u(r, z) = \sum_{n=1}^{\infty} A_n \cosh \alpha_n z J_0(\alpha_n r).$$

From

$$u(r, 4) = 50 = \sum_{n=1}^{\infty} A_n \cosh 4\alpha_n J_0(\alpha_n r)$$

we obtain (as in Example 1 of Section 14.1)

$$A_n \cosh 4\alpha_n = \frac{2(50)}{4J_1^2(2\alpha_n)} \int_0^2 r J_0(\alpha_n r) dr = \frac{50}{\alpha_n J_1(2\alpha_n)}.$$

Thus the temperature in the cylinder is

$$u(r, z) = 50 \sum_{n=1}^{\infty} \frac{\cosh \alpha_n z J_0(\alpha_n r)}{\alpha_n \cosh 4\alpha_n J_1(2\alpha_n)}.$$

14. Using $u = RZ$ and $-\lambda$ as a separation constant and then letting $\lambda = \alpha^2 > 0$ leads to

$$r^2 R'' + rR' + \alpha^2 r^2 R = 0, \quad R'(1) = 0, \quad \text{and} \quad Z'' - \alpha^2 Z = 0.$$

Thus

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)$$

$$Z(z) = c_3 \cosh \alpha z + c_4 \sinh \alpha z$$

for $\alpha > 0$. Arguing that $u(r, z)$ is bounded as $r \rightarrow 0$ we define $c_2 = 0$. Since the eigenvalues are defined by $J'_0(\alpha) = 0$ we know that $\lambda = \alpha = 0$ is an eigenvalue. The solutions are then

$$R(r) = c_1 + c_2 \ln r \quad \text{and} \quad Z(z) = c_3 z + c_4$$

where boundedness again dictates that $c_2 = 0$. Thus,

$$u(r, z) = A_0 z + B_0 + \sum_{n=1}^{\infty} (A_n \sinh \alpha_n z + B_n \cosh \alpha_n z) J_0(\alpha_n r).$$

Finally, the specified conditions $z = 0$ and $z = 1$ give, in turn,

$$B_0 = 2 \int_0^1 r f(r) dr$$

$$B_n = \frac{2}{J_0^2(\alpha_n)} \int_0^1 r f(r) J_0(\alpha_n r) dr$$

$$A_0 = -B_0 + 2 \int_0^1 r g(r) dr$$

$$A_n = \frac{1}{\sinh \alpha_n} \left[-B_n \cosh \alpha_n + \frac{2}{J_0^2(\alpha_n)} \int_0^1 r g(r) J_0(\alpha_n r) dr \right].$$

15. Referring to Example 1 in Section 14.3 of the text we have

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta).$$

For $x = \cos \theta$

$$u(1, \theta) = \begin{cases} 100 & 0 < \theta < \pi/2 \\ -100 & \pi/2 < \theta < \pi \end{cases} = 100 \begin{cases} -1, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases} = g(x).$$

From Problem 22 in Exercise 12.6 we have

$$u(r, \theta) = 100 \left[\frac{3}{2} r P_1(\cos \theta) - \frac{7}{8} r^3 P_3(\cos \theta) + \frac{11}{16} r^5 P_5(\cos \theta) + \cdots \right].$$

16. Since

$$\frac{1}{r} \frac{\partial^2}{\partial r^2}(ru) = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{\partial u}{\partial r} + u \right] = \frac{1}{r} \left[r \frac{\partial^2 u}{\partial r^2} + \frac{\partial u}{\partial r} + \frac{\partial u}{\partial r} \right] = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r}$$

the differential equation becomes

$$\frac{1}{r} \frac{\partial^2}{\partial r^2}(ru) = \frac{\partial^2 u}{\partial t^2} \quad \text{or} \quad \frac{\partial^2}{\partial r^2}(ru) = r \frac{\partial^2 u}{\partial t^2}.$$

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Letting $v(r, t) = ru(r, t)$ we obtain the boundary-value problem

$$\begin{aligned}\frac{\partial^2 v}{\partial r^2} &= \frac{\partial^2 v}{\partial t^2}, & 0 < r < 1, \quad t > 0 \\ \frac{\partial v}{\partial r} \Big|_{r=1} - v(1, t) &= 0, & t > 0 \\ v(r, 0) &= rf(r), \quad \frac{\partial v}{\partial t} \Big|_{t=0} = rg(r), & 0 < r < 1.\end{aligned}$$

If we separate variables using $v(r, t) = R(r)T(t)$ and separation constant $-\lambda$ then we obtain

$$\frac{R''}{R} = \frac{T''}{T} = -\lambda$$

so that

$$\begin{aligned}R'' + \lambda R &= 0 \\ T'' + \lambda T &= 0.\end{aligned}$$

Letting $\lambda = \alpha^2 > 0$ and solving the differential equations we get

$$\begin{aligned}R(r) &= c_1 \cos \alpha r + c_2 \sin \alpha r \\ T(t) &= c_3 \cos \alpha t + c_4 \sin \alpha t.\end{aligned}$$

Since $u(r, t) = v(r, t)/r$, in order to insure boundedness at $r = 0$ we define $c_1 = 0$. Then $R(r) = c_2 \sin \alpha r$. Now the boundary condition $R'(1) - R(1) = 0$ implies $\alpha \cos \alpha - \sin \alpha = 0$. Thus, the eigenvalues λ_n are determined by the positive solutions of $\tan \alpha = \alpha$. We now have

$$v_n(r, t) = (A_n \cos \alpha_n t + B_n \sin \alpha_n t) \sin \alpha_n r.$$

For the eigenvalue $\lambda = 0$,

$$R(r) = c_1 r + c_2 \quad \text{and} \quad T(t) = c_3 t + c_4,$$

and boundedness at $r = 0$ implies $c_2 = 0$. We then take

$$v_0(r, t) = A_0 tr + B_0 r$$

so that

$$v(r, t) = A_0 tr + B_0 r + \sum_{n=1}^{\infty} (a_n \cos \alpha_n t + B_n \sin \alpha_n t) \sin \alpha_n r.$$

Now

$$v(r, 0) = rf(r) = B_0 r + \sum_{n=1}^{\infty} A_n \sin \alpha_n r.$$

Since $\{r, \sin \alpha_n r\}$ is an orthogonal set on $[0, 1]$,

$$\int_0^1 r \sin \alpha_n r \, dr = 0 \quad \text{and} \quad \int_0^1 \sin \alpha_n r \sin \alpha_m r \, dr = 0$$

for $m \neq n$. Therefore

$$\int_0^1 r^2 f(r) \, dr = B_0 \int_0^1 r^2 \, dr = \frac{1}{3} B_0$$

and

$$B_0 = 3 \int_0^1 r^2 f(r) \, dr.$$

Also

$$\int_0^1 r f(r) \sin \alpha_n r \, dr = A_n \int_0^1 \sin^2 \alpha_n r \, dr$$

and

$$A_n = \frac{\int_0^1 r f(r) \sin \alpha_n r \, dr}{\int_0^1 \sin^2 \alpha_n r \, dr}.$$

Now

$$\int_0^1 \sin^2 \alpha_n r \, dr = \frac{1}{2} \int_0^1 (1 - \cos 2\alpha_n r) \, dr = \frac{1}{2} \left[1 - \frac{\sin 2\alpha_n}{2\alpha_n} \right] = \frac{1}{2} [1 - \cos^2 \alpha_n].$$

Since $\tan \alpha_n = \alpha_n$,

$$1 + \alpha_n^2 = 1 + \tan^2 \alpha_n = \sec^2 \alpha_n = \frac{1}{\cos^2 \alpha_n}$$

and

$$\cos^2 \alpha_n = \frac{1}{1 + \alpha_n^2}.$$

Then

$$\int_0^1 \sin^2 \alpha_n r \, dr = \frac{1}{2} \left[1 - \frac{1}{1 + \alpha_n^2} \right] = \frac{\alpha_n^2}{2(1 + \alpha_n^2)}$$

and

$$A_n = \frac{2(1 + \alpha_n^2)}{\alpha_n^2} \int_0^1 r f(r) \sin \alpha_n r \, dr.$$

Similarly, setting

$$\left. \frac{\partial v}{\partial t} \right|_{t=0} = r g(r) = A_0 r + \sum_{n=1}^{\infty} B_n \alpha_n \sin \alpha_n r$$

we obtain

$$A_0 = 3 \int_0^1 r^2 g(r) \, dr$$

and

$$B_n = \frac{2(1 + \alpha_n^2)}{\alpha_n^3} \int_0^1 r g(r) \sin \alpha_n r \, dr.$$

Therefore, since $v(r, t) = ru(r, t)$ we have

$$u(r, t) = A_0 t + B_0 + \sum_{n=1}^{\infty} (A_n \cos \alpha_n t + B_n \sin \alpha_n t) \frac{\sin \alpha_n r}{r},$$

where the α_n are solutions of $\tan \alpha = \alpha$ and

$$A_0 = 3 \int_0^1 r^2 g(r) \, dr$$

$$B_0 = 3 \int_0^1 r^2 f(r) \, dr$$

$$A_n = \frac{2(1 + \alpha_n^2)}{\alpha_n^2} \int_0^1 r f(r) \sin \alpha_n r \, dr$$

$$B_n = \frac{2(1 + \alpha_n^2)}{\alpha_n^3} \int_0^1 r g(r) \sin \alpha_n r \, dr$$

for $n = 1, 2, 3, \dots$

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17. We note that the differential equation can be expressed in the form

$$\frac{d}{dx}[xu'] = -\alpha^2 xu.$$

Thus

$$u_n \frac{d}{dx}[xu'_m] = -\alpha_m^2 xu_m u_n$$

and

$$u_m \frac{d}{dx}[xu'_n] = -\alpha_n^2 xu_n u_m.$$

Subtracting we obtain

$$u_n \frac{d}{dx}[xu'_m] - u_m \frac{d}{dx}[xu'_n] = (\alpha_n^2 - \alpha_m^2) xu_m u_n$$

and

$$\int_a^b u_n \frac{d}{dx}[xu'_m] dx - \int_a^b u_m \frac{d}{dx}[xu'_n] dx = (\alpha_n^2 - \alpha_m^2) \int_a^b xu_m u_n dx.$$

Using integration by parts this becomes

$$\begin{aligned} u_n xu'_m \Big|_a^b - \int_a^b xu'_m u'_n dx - u_m xu'_n \Big|_a^b + \int_a^b xu'_n u'_m dx \\ = b[u_n(b)u'_m(b) - u_m(b)u'_n(b)] - a[u_n(a)u'_m(a) - u_m(a)u'_n(a)] \\ = (\alpha_n^2 - \alpha_m^2) \int_a^b xu_m u_n dx. \end{aligned}$$

Since

$$u(x) = Y_0(\alpha a)J_0(\alpha x) - J_0(\alpha a)Y_0(\alpha x)$$

we have

$$u_n(b) = Y_0(\alpha_n a)J_0(\alpha_n b) - J_0(\alpha_n a)Y_0(\alpha_n b) = 0$$

by the definition of the α_n . Similarly $u_m(b) = 0$. Also

$$u_n(a) = Y_0(\alpha a)J_0(\alpha_n a) - J_0(\alpha_n a)Y_0(\alpha a) = 0$$

and $u_m(a) = 0$. Therefore

$$\int_a^b xu_m u_n dx = \frac{1}{\alpha_n^2 - \alpha_m^2} (b[u_n(b)u'_m(b) - u_m(b)u'_n(b)] - a[u_n(a)u'_m(a) - u_m(a)u'_n(a)]) = 0$$

and the $u_n(x)$ are orthogonal with respect to the weight function x .

18. Letting $u(r, t) = R(r)T(t)$ and the separation constant be $-\lambda = -\alpha^2$ we obtain

$$rR'' + R' + \alpha^2 rR = 0$$

$$T' + \alpha^2 T = 0,$$

with solutions

$$R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)$$

$$T(t) = c_3 e^{-\alpha^2 t}.$$

Now the boundary conditions imply

$$R(a) = 0 = c_1 J_0(\alpha a) + c_2 Y_0(\alpha a)$$

$$R(b) = 0 = c_1 J_0(\alpha b) + c_2 Y_0(\alpha b)$$

so that

$$c_2 = -\frac{c_1 J_0(\alpha a)}{Y_0(\alpha a)}$$

and

$$c_1 J_0(\alpha b) - \frac{c_1 J_0(\alpha a)}{Y_0(\alpha a)} Y_0(\alpha b) = 0$$

or

$$Y_0(\alpha a) J_0(\alpha b) - J_0(\alpha a) Y_0(\alpha b) = 0.$$

This equation defines α_n for $n = 1, 2, 3, \dots$. Now

$$R(r) = c_1 J_0(\alpha r) - c_1 \frac{J_0(\alpha a)}{Y_0(\alpha a)} Y_0(\alpha r) = \frac{c_1}{Y_0(\alpha a)} [Y_0(\alpha a) J_0(\alpha r) - J_0(\alpha a) Y_0(\alpha r)]$$

and

$$u_n(r, t) = A_n [Y_0(\alpha_n a) J_0(\alpha_n r) - J_0(\alpha_n a) Y_0(\alpha_n r)] e^{-\alpha_n^2 t} = A_n u_n(r) e^{-\alpha_n^2 t}.$$

Thus

$$u(r, t) = \sum_{n=1}^{\infty} A_n u_n(r) e^{-\alpha_n^2 t}.$$

From the initial condition

$$u(r, 0) = f(r) = \sum_{n=1}^{\infty} A_n u_n(r)$$

we obtain

$$A_n = \frac{\int_a^b r f(r) u_n(r) dr}{\int_a^b r u_n^2(r) dr}.$$

19. We use the superposition principle for Laplace's equation discussed in Section 13.5 and shown schematically in Figure 13.15 in the text. That is,

$$\text{Solution } u = \text{Solution } u_1 \text{ of Problem 1} + \text{Solution } u_2 \text{ of Problem 2,}$$

where in Problem 1 the boundary condition on the top and bottom of the cylinder is $u = 0$, while on the lateral surface $r = c$ it is $u = h(z)$, and in Problem 2 the boundary condition on the top of the cylinder $z = L$ is $u = f(r)$, on the bottom $z = 0$ it is $u = g(r)$, and on the lateral surface $r = c$ it is $u = 0$.

20. Solution for $u_1(r, z)$

Using λ as a separation constant we have

$$\frac{R'' + \frac{1}{r}R'}{R} = -\frac{Z''}{Z} = \lambda,$$

so

$$rR'' + R' - \lambda rR = 0 \quad \text{and} \quad Z'' + \lambda Z = 0.$$

The differential equation in Z , together with the boundary conditions $Z(0) = 0$ and $Z(L) = 0$ is a Sturm-Liouville problem. Letting $\lambda = \alpha^2 > 0$ we note that the above differential equation in R is a modified parametric Bessel equation which is discussed in Section 5.3 in the text. Also, we have $Z(z) = c_1 \cos \alpha z + c_2 \sin \alpha z$. The boundary conditions imply $c_1 = 0$ and $\sin \alpha L = 0$. Thus, $\alpha_n = n\pi/L$, $n = 1, 2, 3, \dots$, so $\lambda_n = n^2\pi^2/L^2$ and

$$R(r) = c_3 I_0\left(\frac{n\pi}{L}r\right) + c_4 K_0\left(\frac{n\pi}{L}r\right).$$

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Now boundedness at $r = 0$ implies $c_4 = 0$, so $R(r) = c_3 I_0(n\pi r/L)$ and

$$u_1(r, z) = \sum_{n=1}^{\infty} A_n I_0\left(\frac{n\pi}{L}r\right) \sin\left(\frac{n\pi}{L}z\right).$$

At $r = c$ for $0 < z < L$ we have

$$h(z) = u_1(c, z) = \sum_{n=1}^{\infty} A_n I_0\left(\frac{n\pi}{L}c\right) \sin\left(\frac{n\pi}{L}z\right)$$

which gives

$$A_n = \frac{2}{LI_0(n\pi c/L)} \int_0^L h(z) \sin\left(\frac{n\pi}{L}z\right) dz.$$

Solution for $u_2(r, z)$

In this case we use $-\lambda$ as a separation constant which leads to

$$\frac{R'' + \frac{1}{r}R'}{R} = -\frac{Z''}{Z} = -\lambda,$$

so

$$rR'' + R' + \lambda rR = 0 \quad \text{and} \quad Z'' - \lambda Z = 0.$$

The differential equation in R is a parametric Bessel equation. Using $\lambda = \alpha^2$ we find $R(r) = c_1 J_0(\alpha r) + c_2 Y_0(\alpha r)$. Boundedness at $r = 0$ implies $c_2 = 0$ so $R(r) = c_3 J_0(\alpha r)$. The boundary condition $R(c) = 0$ then gives the defining equation for the eigenvalues: $J_0(\alpha c) = 0$. Let $\lambda_n = \alpha_n^2$ where $\alpha_n c = x_n$ are the roots. The solution of the differential equation in Z is $Z(z) = c_4 \cosh \alpha_n z + c_5 \sinh \alpha_n z$, so

$$u_2(r, z) = \sum_{n=1}^{\infty} (B_n \cosh \alpha_n z + C_n \sinh \alpha_n z) J_0(\alpha_n r).$$

At $z = 0$, for $0 < r < c$, we have

$$f(r) = u_2(r, 0) = \sum_{n=1}^{\infty} B_n J_0(\alpha_n r),$$

so

$$B_n = \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^c r f(r) J_0(\alpha_n r) dr.$$

At $z = L$, for $0 < r < c$, we have

$$g(r) = u_2(r, L) = \sum_{n=1}^{\infty} (B_n \cosh \alpha_n L + C_n \sinh \alpha_n L) J_0(\alpha_n r),$$

so

$$B_n \cosh \alpha_n L + C_n \sinh \alpha_n L = \frac{2}{c^2 J_1^2(\alpha_n c)} \int_0^c r g(r) J_0(\alpha_n r) dr$$

and

$$C_n = -B_n \frac{\cosh \alpha_n L}{\sinh \alpha_n L} + \frac{2}{c^2 (\sinh \alpha_n L) J_1^2(\alpha_n c)} \int_0^c r g(r) J_0(\alpha_n r) dr.$$

By the superposition principle the solution of the original problem is

$$u(r, z) = u_1(r, z) + u_2(r, z).$$